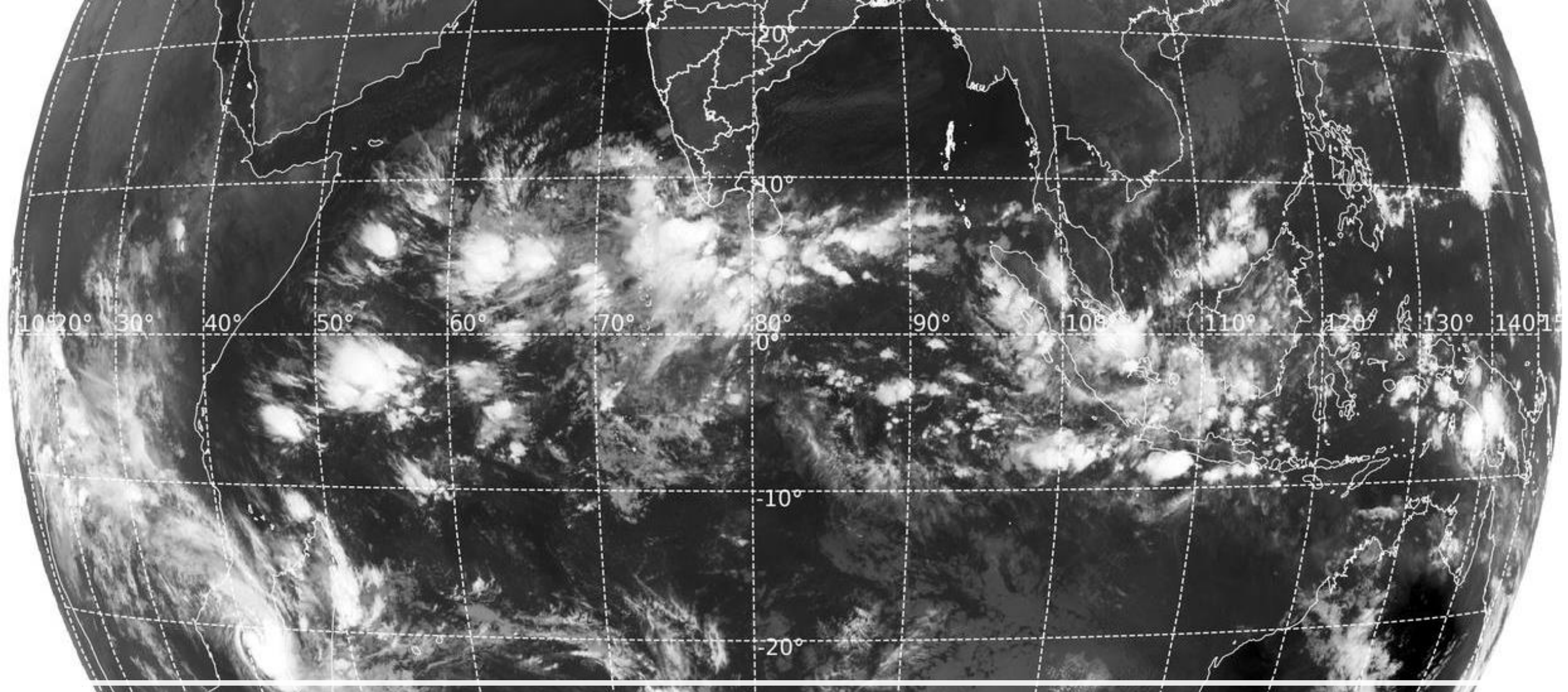




Remote Sensing and Geographic Information Science

Note: In this ppt, figures/text are adopted from various sources; this is only for class explanation purpose.



Digital Image Processing Image Enhancement

(i) Concept of Color and Color Composites

Objectives

- **Color Fundamentals**
- **Chromaticity Diagram**
- **Color Models**
- **Color Composites**

Color Fundamentals

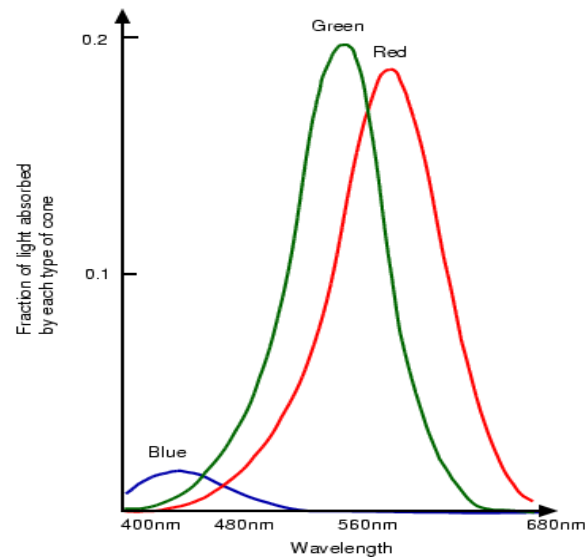
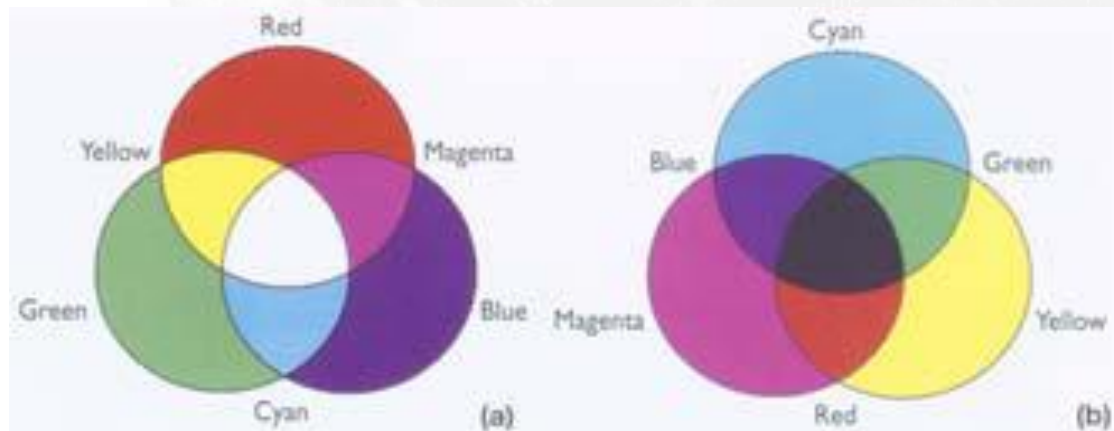


Figure : Spectral response curves for each cone type.
The peaks for each curve are at 440nm (blue),
545nm (green) and 580nm (red).

- In eyes, cones are responsible for color perception.
- Of the 6-7 million cones of human eye:
 - 65% are sensitive to red light
 - 33% to green light
 - 2% to blue light
- These form the primary colors

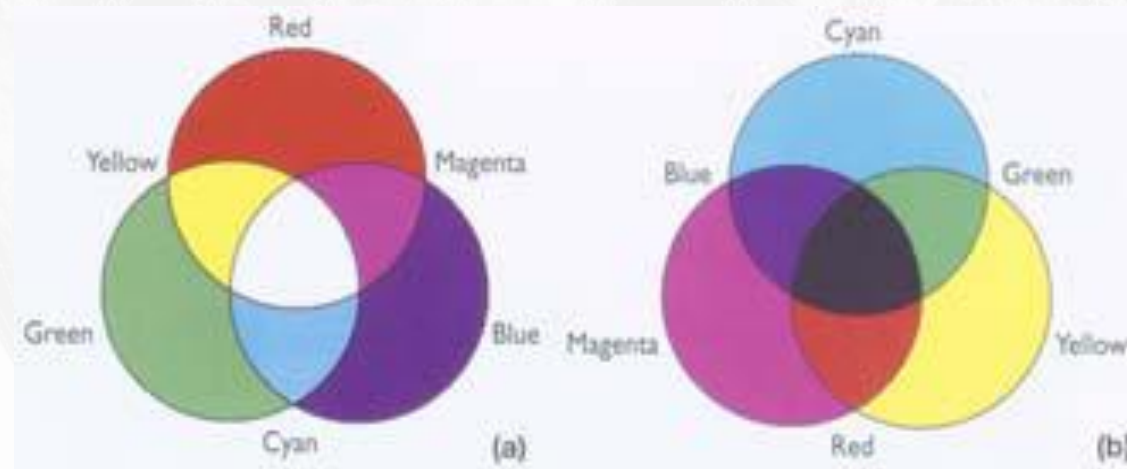
Color Fundamentals...

- Primary Additive colours
: Red, Green and Blue (RGB)
- Complementary Colours
: Cyan, Magenta and Yellow



Color Fundamentals- Natural Color Photography

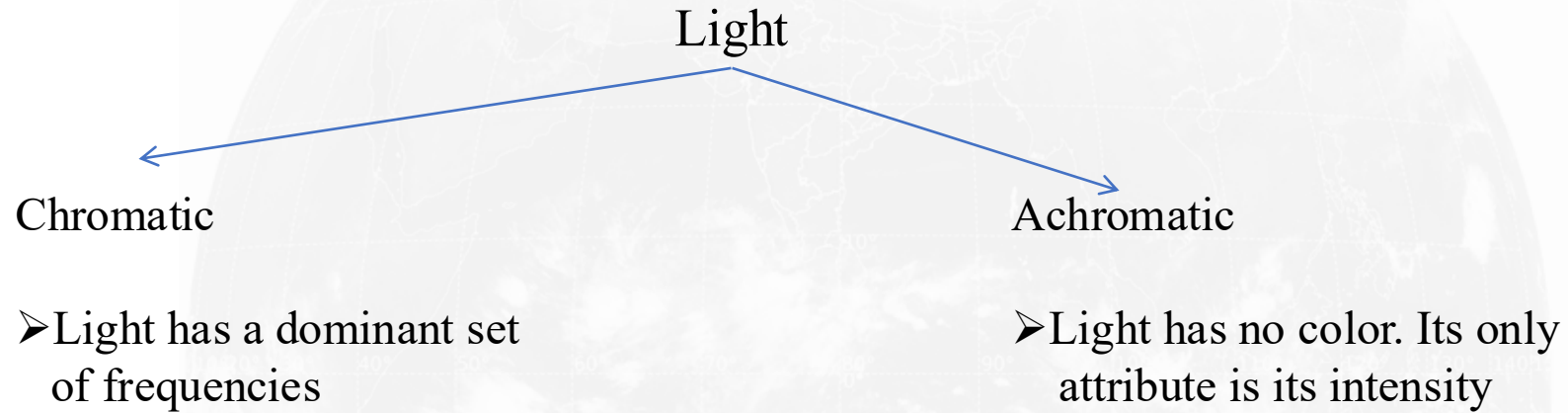
- If three primary colors are superimposed in unequal amounts, then number of colors are produced
- If three primary colors are superimposed in equal amounts, then greys ranging from black to white are produced.
- If white light is passed through a color filter, it is possible to subtract one of the primary colors



Additive process

Subtractive process

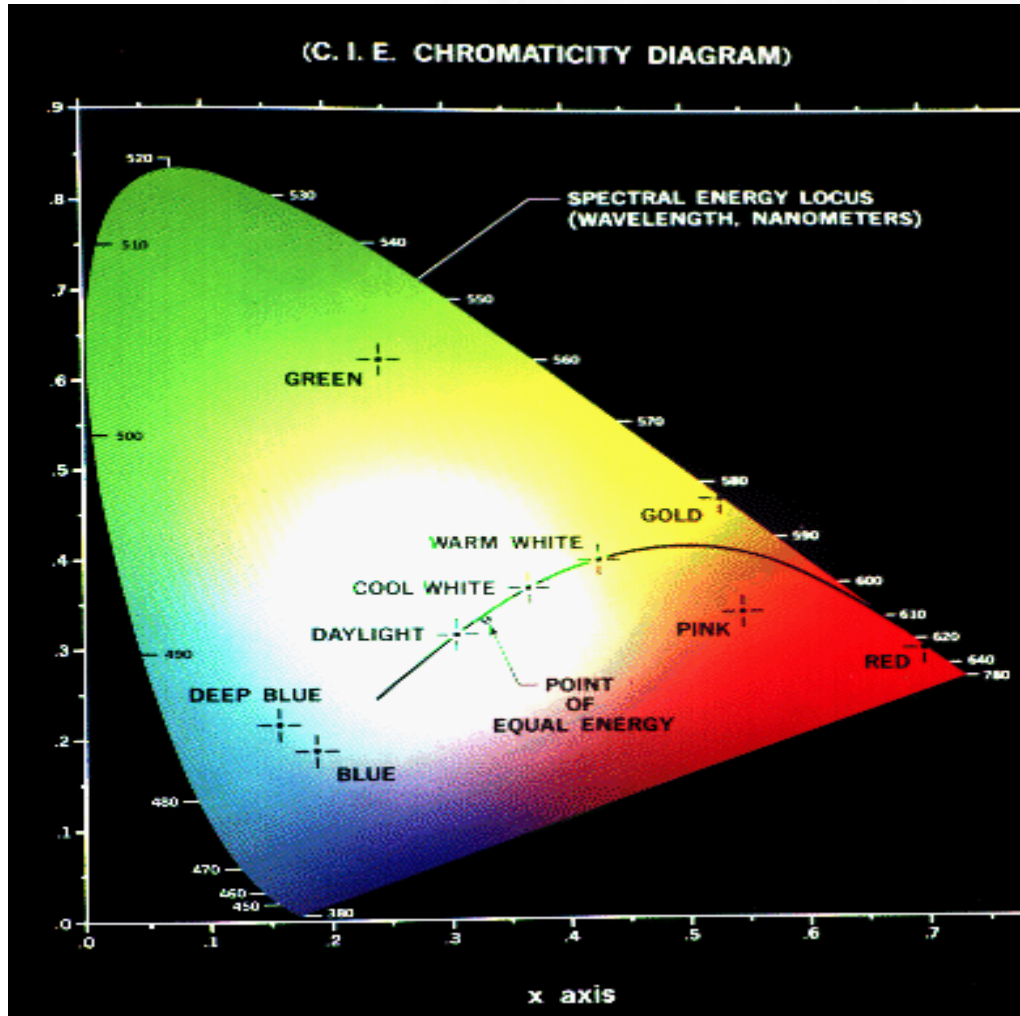
Color Fundamentals...



Color models

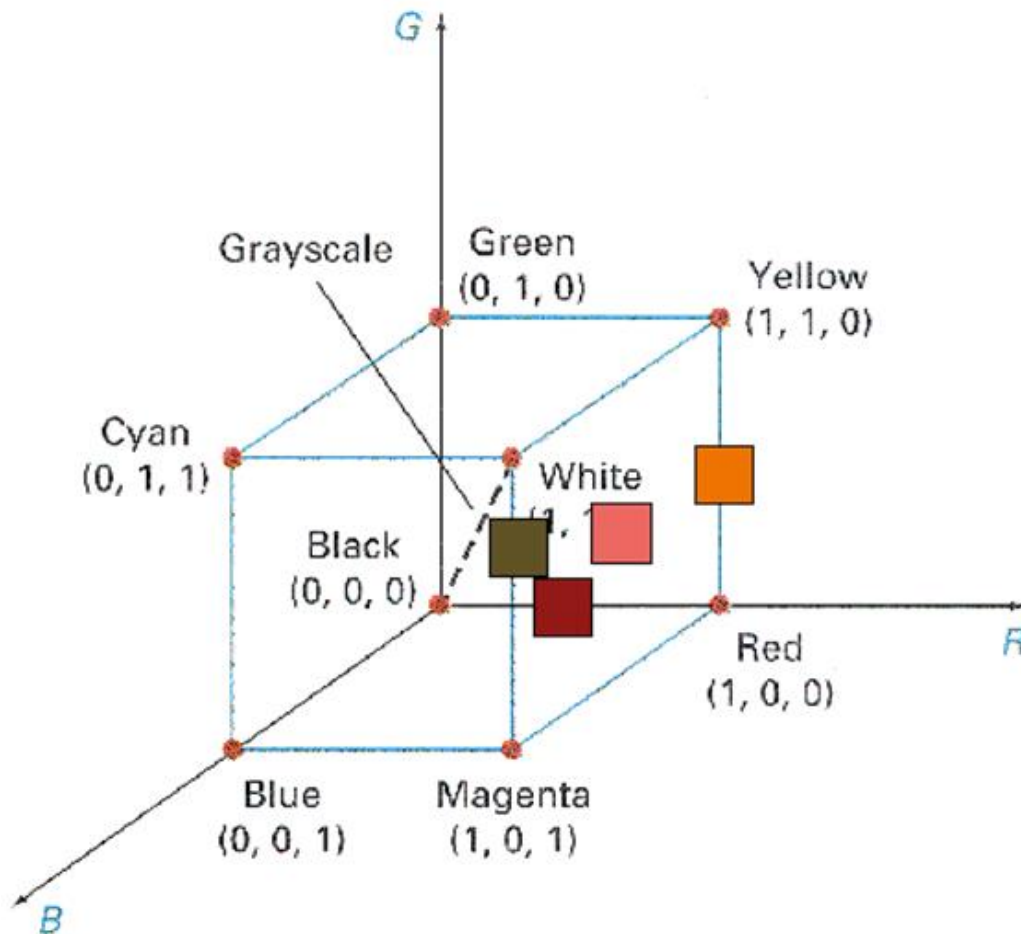
- Standard means to specify colors by defining a 3D coordinate system.
- The sub space will contain all possible color combinations within a particular model
- E.g.: RGB, CMY, IHS.

Chromaticity Diagram



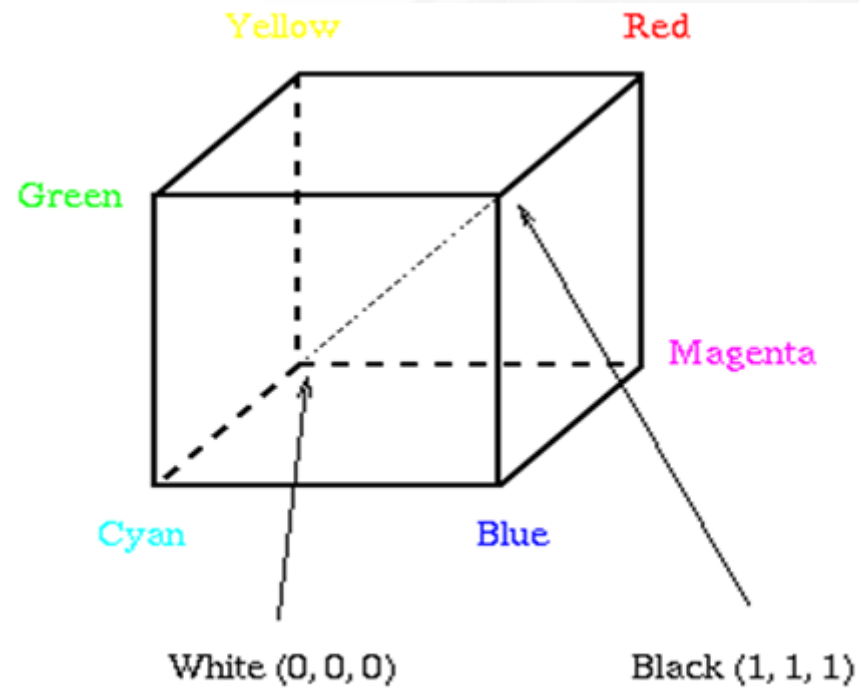
- It is often convenient to work in a 2D color space.
- Chromaticity diagrams show color composites as a function of x (red), y (green) and z (which is $1-x-y$).
- Devices such as colorimeter measures color using numbers derived from CIE values

Color Space - RGB



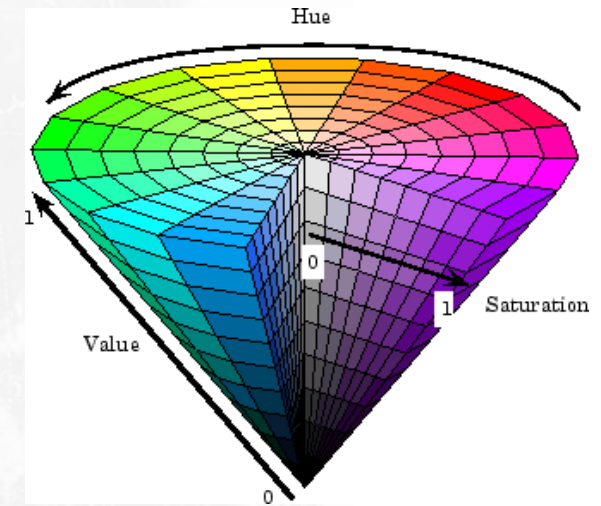
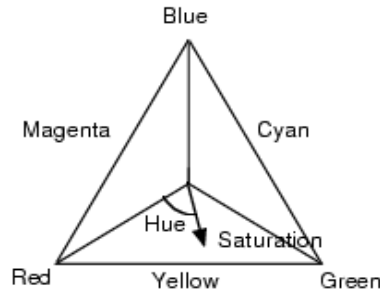
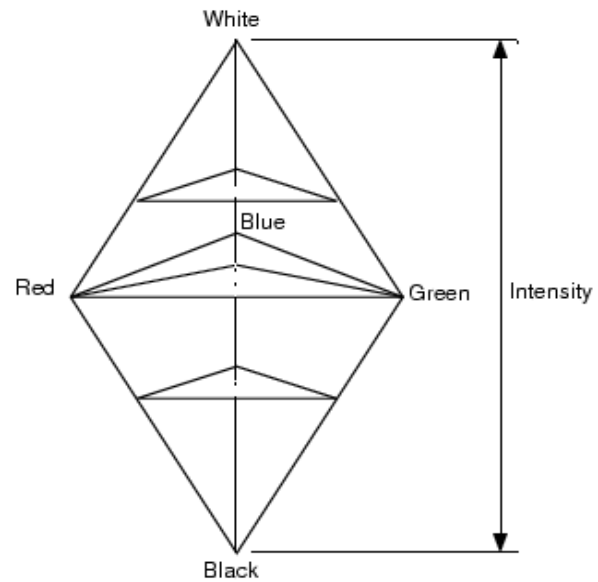
- Primary colors of red, green and blue are used within a cartesian coordinate system
- A unit cube is shown with the underlying assumption that all colors are normalized.

Color Space - CMY



- Cyan (C), Magenta (M) and Yellow (Y) comprise the secondary colors of light.
- This color space is generally used to generate hardcopy output.

Color Space-IHS



Source: www.mathworks.co.uk

- Humans define color in terms of its intensity (I), hue (H) and saturation (S).
- Intensity: Variations in brightness
- Hue: Dominant wavelength of color
- Saturation: Purity of color

Photographic Remote Sensing

Multispectral

- Involves simultaneously obtaining images on the same scene at different wavelengths
- Four: Blue, Green, Red and NIR parts of EMR
- Multispectral imaging allows the examination of single band images
- Natural and False colour composites can be produced

Color Composites

True Colour Composite (TCC)

- Red band – Red; Green band – Green; Blue band – Blue

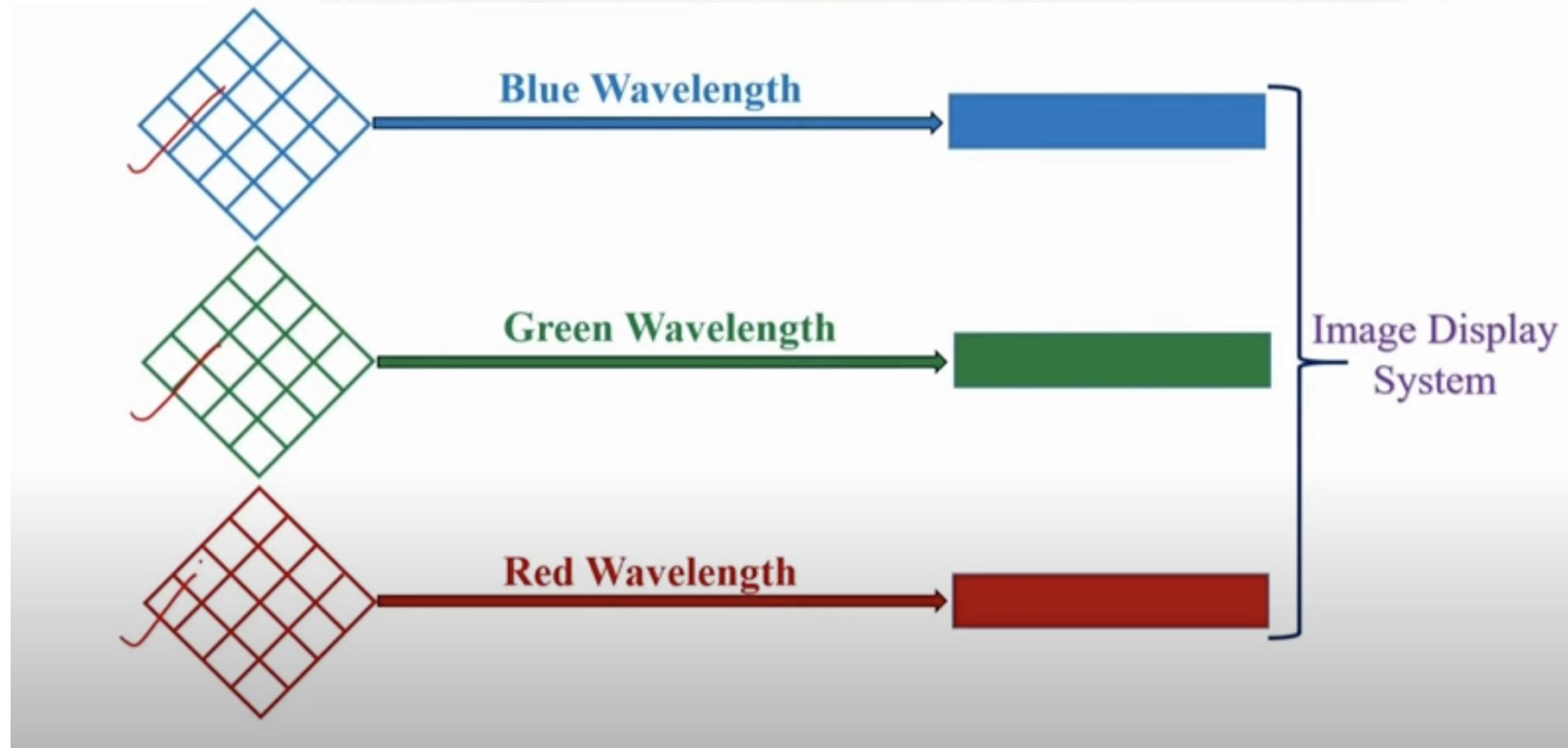
False Colour Composite (FCC)

- Any other combination of colours
- E.g., Blue band – Red; Red band – Green; Green band – Blue
- E.g., Blue band – Red; Red band – Green; NIR band – Blue

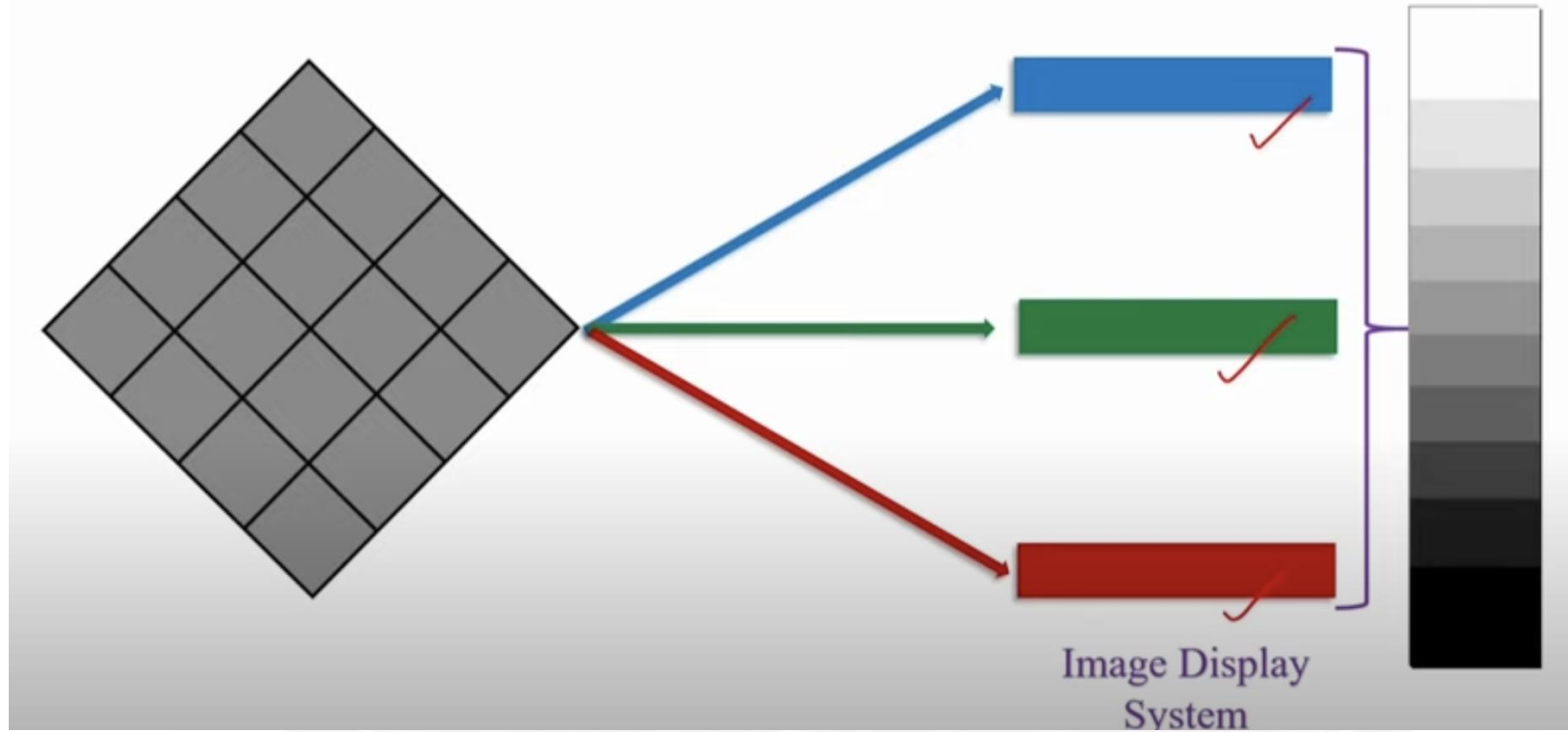
Standard False Colour Composite (FCC)

- E.g., NIR band – Red; Red band – Green; Green band – Blue
- In IRS: Band 4 – Red; Band 3 – Green; Band 2 – Blue

Natural Color Composites



Grey Scale Image



Color Composites

Landsat TM

Average Orbital Height: 700 km (440 Miles)

Spatial Resolution: 30 m, except band 6 which is 90 m

Records Data in 7 Wavelength Intervals (bands)

1. Visible Blue (0.45 to 0.52 microns)
2. Visible Green (0.52 to 0.60 microns)
3. Visible Red (0.63 to 0.69 microns)
4. Near Infrared (0.76 to 0.90 microns)
5. Mid Infrared (1.55 to 1.75 microns)
6. Thermal Infrared (10.4 to 12.5 microns)
7. Mid Infrared (2.08 to 2.35 microns)

Bands 1,2,3,4,5, and 7 record reflected energy

Band 6 records emitted thermal (heat) energy

LANDSAT – SPECTRAL BANDS

Landsat 7			Landsat 8		
<u>Band Name</u>	<u>Bandwidth (μm)</u>	<u>Resolution (m)</u>	<u>Band Name</u>	<u>Bandwidth (μm)</u>	<u>Resolution (m)</u>
			Band 1 Coastal	0.43 – 0.45	30
Band 1 Blue	0.45 – 0.52	30	Band 2 Blue	0.45 – 0.51	30
Band 2 Green	0.52 – 0.60	30	Band 3 Green	0.53 – 0.59	30
Band 3 Red	0.63 – 0.69	30	Band 4 Red	0.64 – 0.67	30
Band 4 NIR	0.77 – 0.90	30	Band 5 NIR	0.85 – 0.88	30
Band 5 SWIR 1	1.55 – 1.75	30	Band 6 SWIR 1	1.57 – 1.65	30
Band 7 SWIR 2	2.09 – 2.35	30	Band 7 SWIR 2	2.11 – 2.29	30
Band 8 Pan	0.52 – 0.90	15	Band 8 Pan	0.50 – 0.68	15
			Band 9 Cirrus	1.36 – 1.38	30
Band 6 TIR	10.40 – 12.50	30/60	Band 10 TIRS 1	10.6 – 11.19	100
			Band 11 TIRS 2	11.5 – 12.51	100

Optimum Index Factor

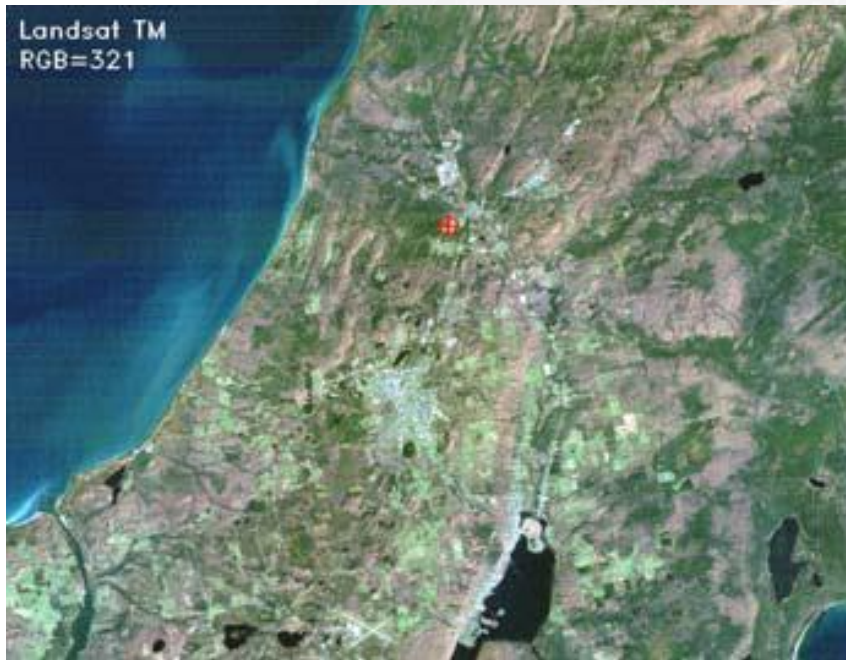
➤ When satellites like Thematic Mapper (TM) are capable of generating more than one color composite, OIF enables to select the best combination.

➤ OIF is given by the expression:

$$OIF = \frac{\sum_{K=1}^3 S_K}{\sum_{J=1}^3 Abs(R_J)}$$

Where S_K denotes the standard deviation for band K and R_J denotes the absolute value of the correlation coefficient between any two of the three bands which are being evaluated.

The Optimum Index Factor (OIF) determines the three-band combination that maximizes the variability (information) in a multi-spectral scene. The index is a ratio of the total variance (standard deviation) within and the correlation between all possible band combinations. The bands that comprise the highest scoring combination are used as the three color channels required for composite.



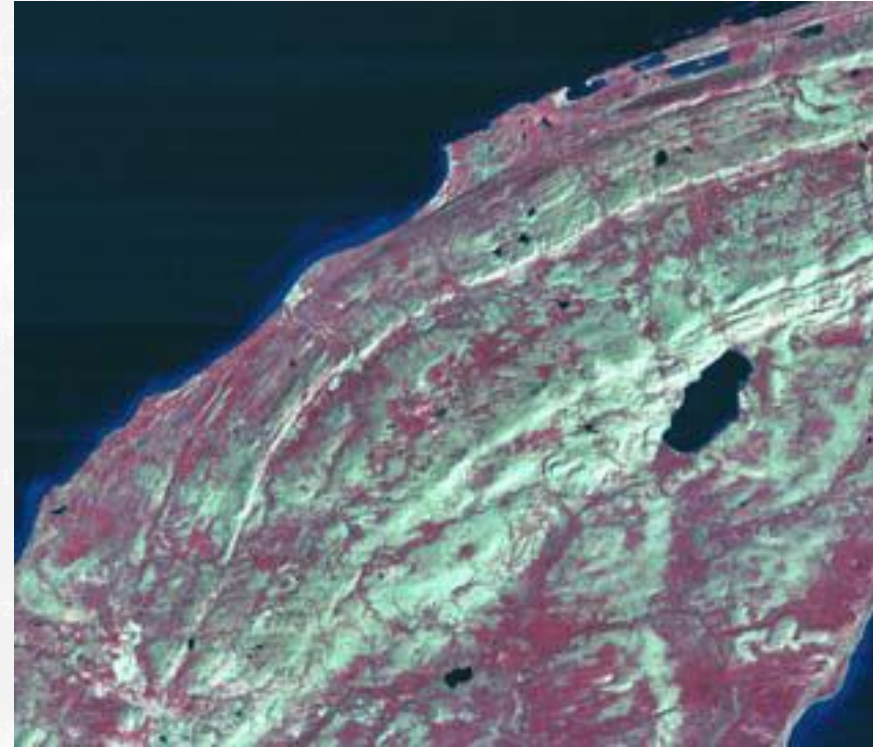
Band 321 (TCC)



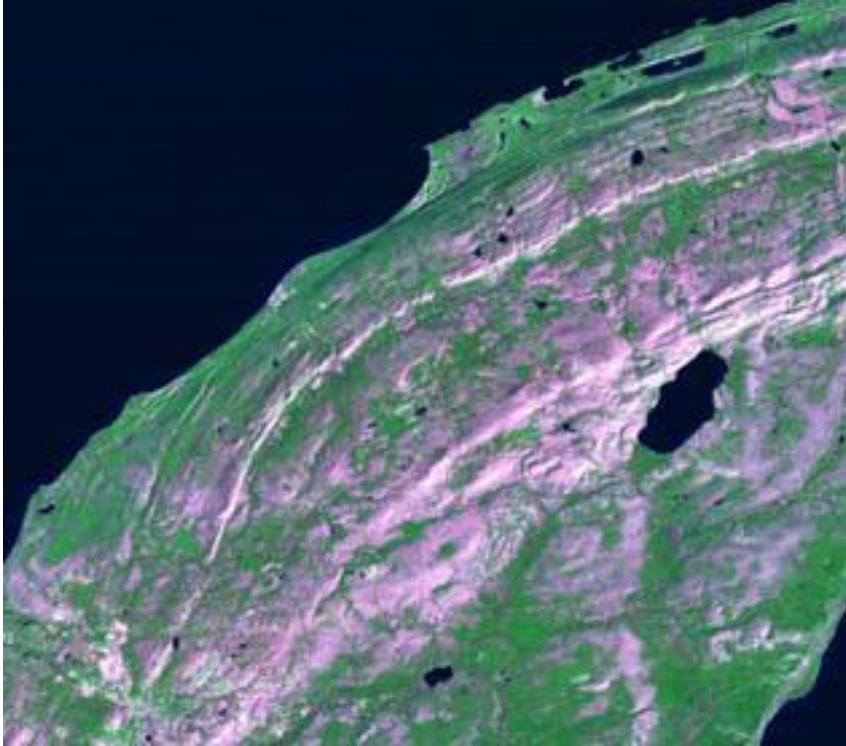
FCC (Band 543)



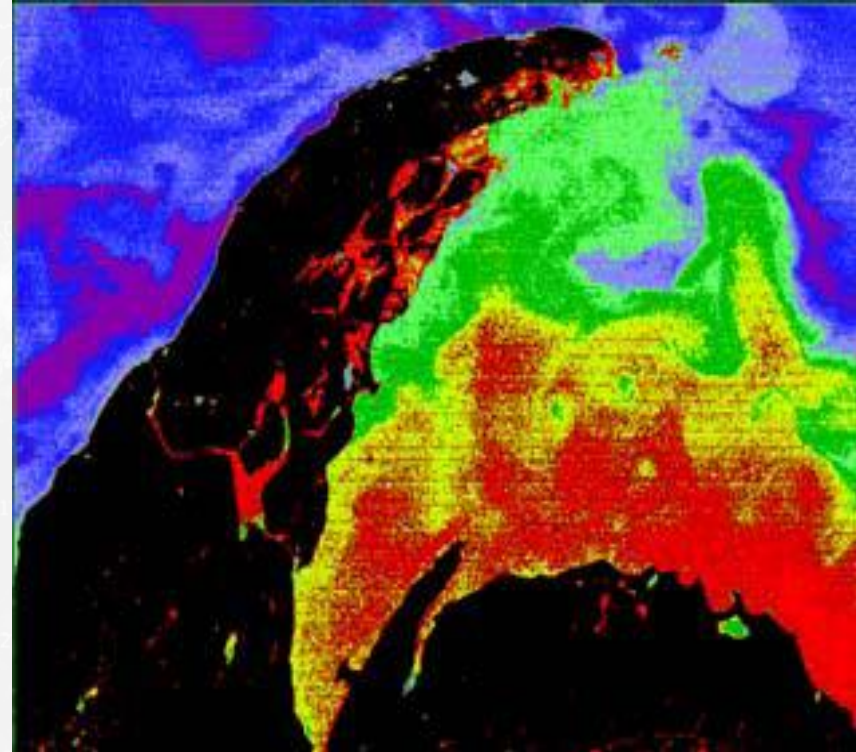
Band 321 (TCC)



Standard FCC (Bands 432)



Band 543 (FCC)

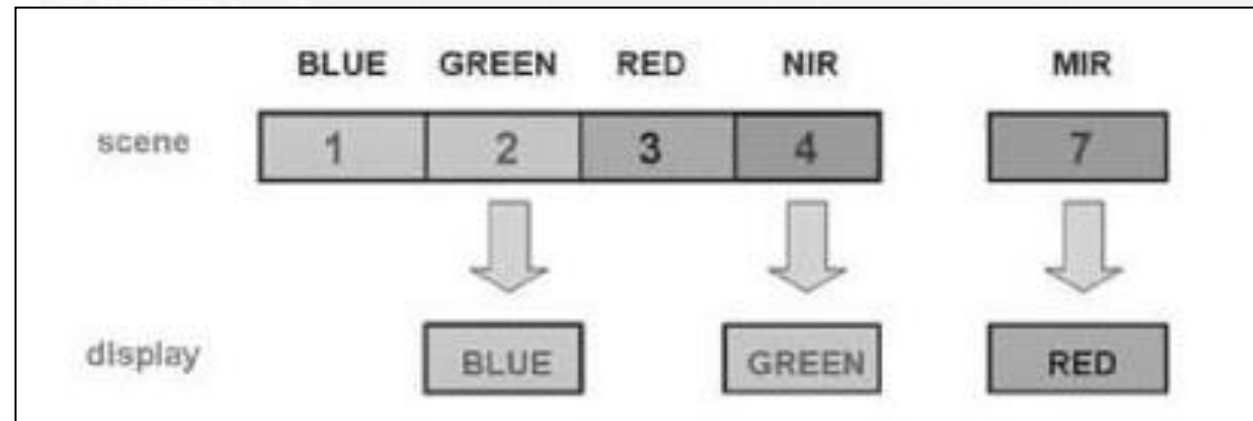


Temperature Image

LANDSAT – BAND COMBINATIONS

742 Band Combination

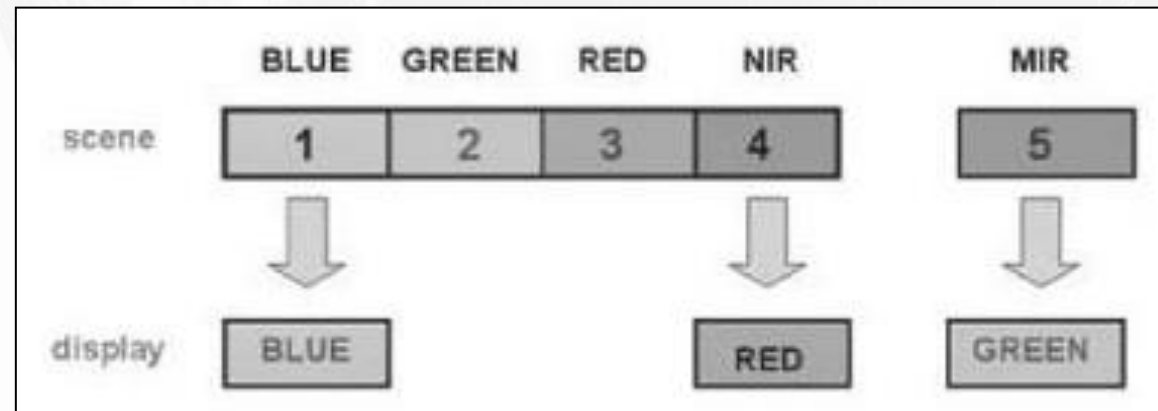
- ✓ Uses one region from visible, one from NIR and one from MIR bands
- ✓ Uses False colours, such that they resembles natural appearance in respective bands
- ✓ Vegetation – Green; Barren land – Pink; Dry / sparse vegetation – orange and brown; Open water – Blue
- ✓ Employed for geologic and agricultural analysis



LANDSAT – BAND COMBINATIONS

451 Band Combination

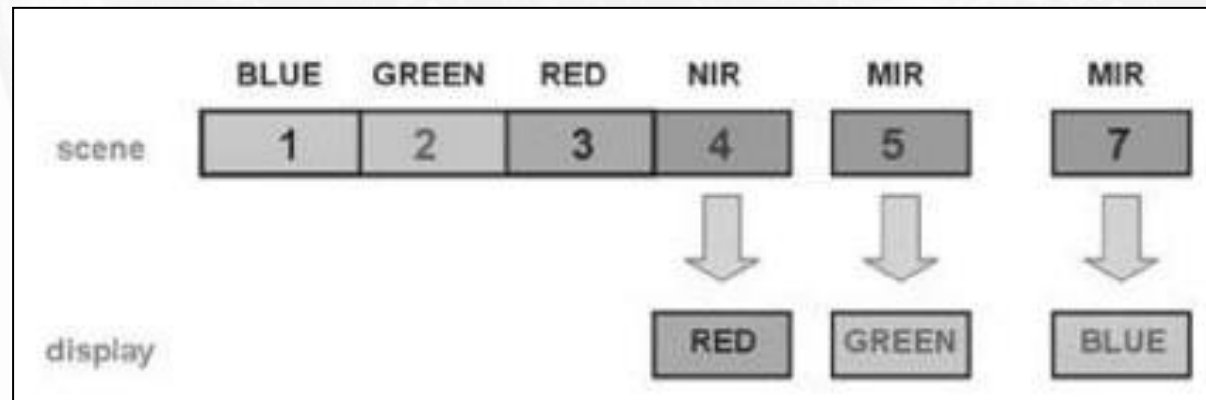
- ✓ Uses Blue, NIR and MIR bands
- ✓ Deep and clear water bodies appear very dark
- ✓ Shallow and turbid water appears as shades of lighter blue
- ✓ Healthy vegetation is represented in red, brown, orange
- ✓ Urban features are represented in white, cyan, and grey
- ✓ Bare soil appear as green and brown



LANDSAT – BAND COMBINATIONS

754 Band Combination

- ✓ Uses three bands from outside of visible region
- ✓ Used for geological analysis
- ✓ It is free from effects of atmospheric scattering (long wave)
- ✓ Coastlines are clearly and sharply defined



LANDSAT – BAND COMBINATIONS

543 Band Combination

- ✓ Uses NIR, MIR, and RED regions of spectrum
- ✓ Edges of water bodies are sharply defined
- ✓ Variation in vegetation type and status is clearly represented in brown, green, and orange
- ✓ Sensitive to variation in soil moisture
- ✓ Useful for soil and vegetation analysis

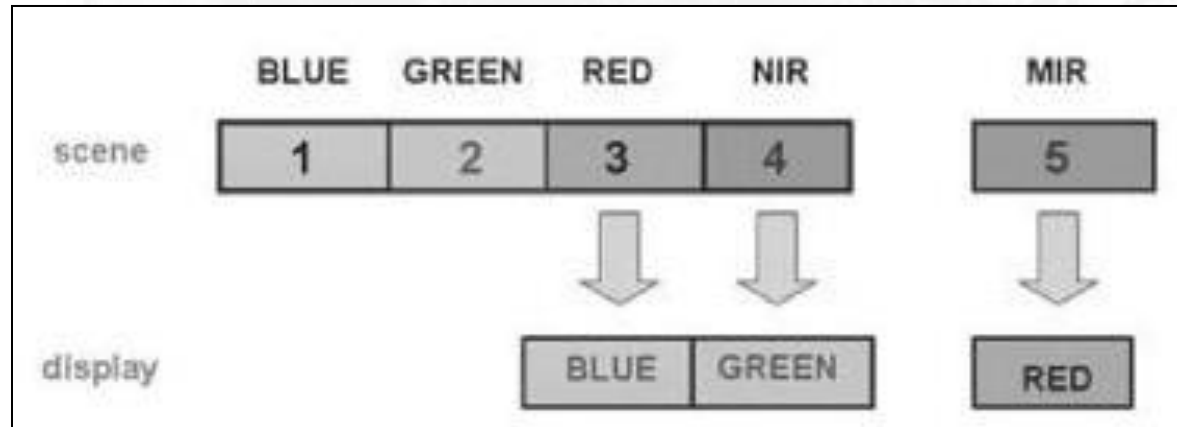


IMAGE INTERPRETATION

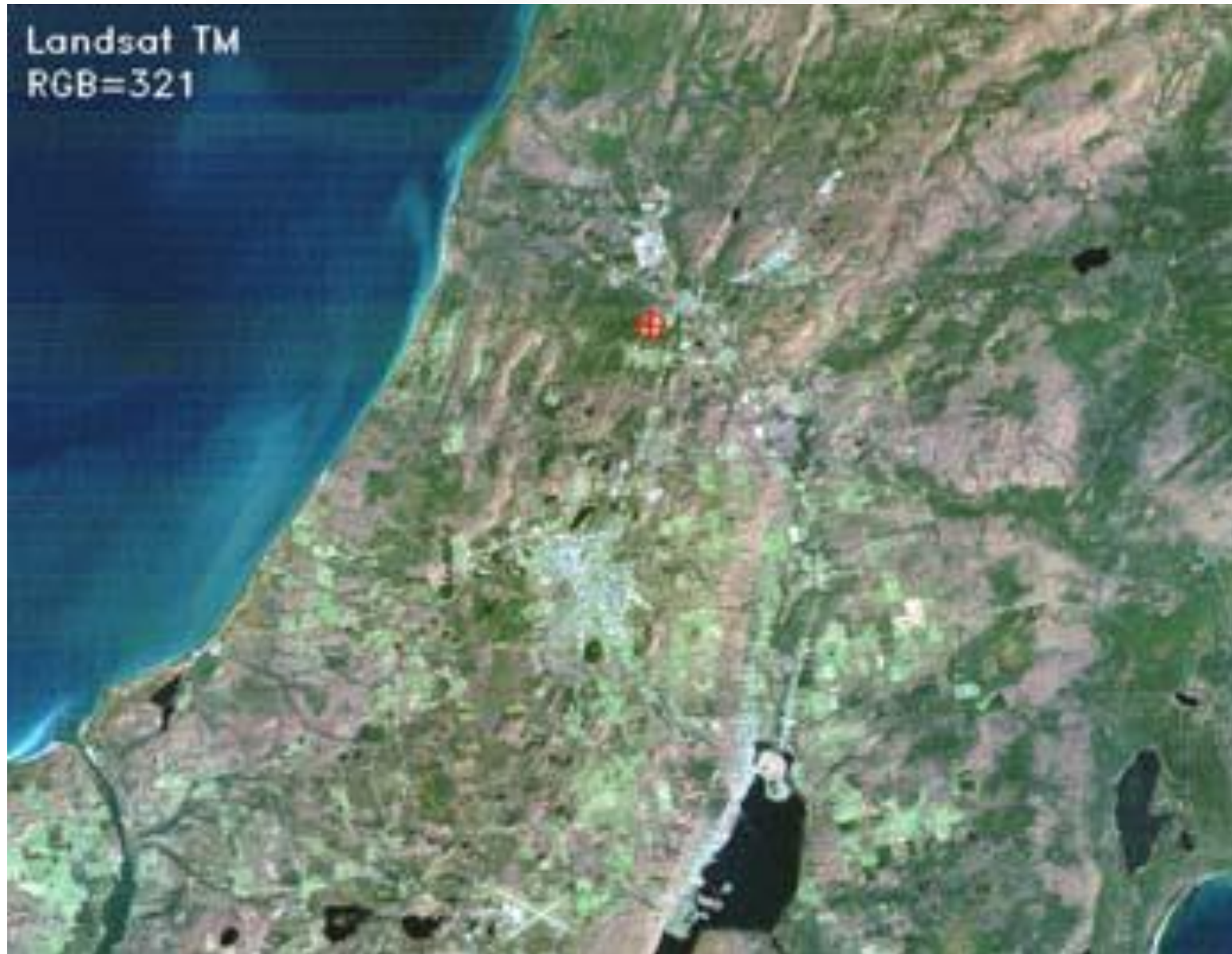


IMAGE INTERPRETATION

- Images of the Earth you see on the computer are taken from cameras mounted on aircraft or satellites in space
- Images from aircraft are called aerial photographs
- Images from satellites are called satellite imagery
- Maps are representational drawings of Earth features while images are actual pictures of the Earth
- Ex: We may be able to see roads on an image, but are not labeled as roads. Image interpretation helps in identifying the image features for further use
- Image interpretation is the act of examining photographic (or, satellite) images for the purpose of identifying objects and judging their significance
- Ex: Identification and classification of structural faults and lineaments from a satellite image

IMAGE INTERPRETATION

- Image interpretation is a method (process) of deriving useful information from the raw image
- Image interpretation requires
 - Knowledge of the subject of interest (and associated fields)
 - Knowledge of geographic region of interest
 - Knowledge of remote sensing system
- Remote Sensing based Image Interpretation differs from regular (human eye / photograph) interpretation as:
 - Images are portrayed from overhead view
 - Images use radiation outside Visible Region (most generally)
 - Images are taken at unfamiliar scales and resolutions

IMAGE INTERPRETATION

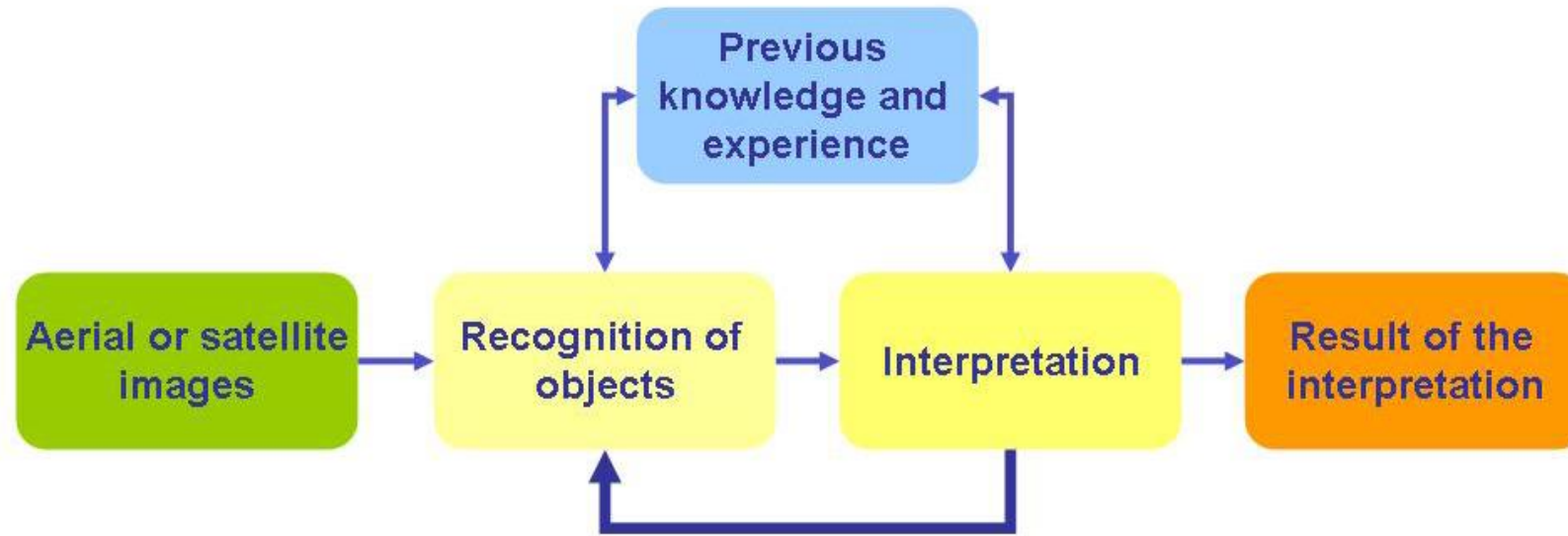


IMAGE INTERPRETATION

- Image interpretation Elements

- Image tone ; Image texture ; Shadow ; Pattern ; Association ; Shape ; Size ; Site

- Image interpretation Tasks

- Classification ; Detection ; Recognition ; Identification ; Enumeration ; Mensuration ; Delineation

IMAGE INTERPRETATION ELEMENTS

IMAGE TONE (COLOUR)

- Denotes the lightness or darkness of a region within an image
- For B/W Images: Light; Medium Gray; Dark Gray; Dark
- For Colour / CIR Image: Tone simply refers to colour (Pale Green; Dark Blue;)
- Image tone is influenced by Intensity of Radiation; Angle of Illumination; and Processing of Films

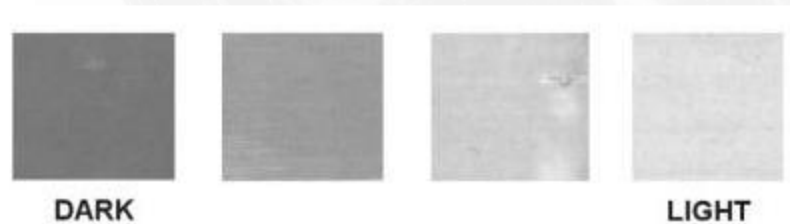


IMAGE INTERPRETATION ELEMENTS

IMAGE TEXTURE

- Refers to the apparent roughness or smoothness of an image region
- Caused by pattern of highlighted and shadowed areas created when an irregular surface is illuminated from an oblique angle
- Ex: A mature forest is rough in texture where as a mature wheat crop is smooth in texture
- Image texture depends on: Nature of surface; Angle of illumination

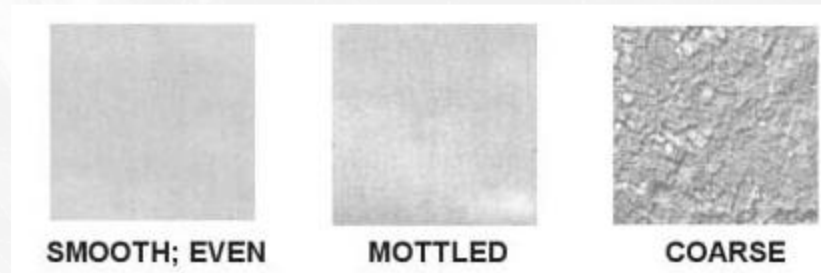


IMAGE INTERPRETATION ELEMENTS

SHADOW

- An object when illuminated at an angle, casts a shadow that may reveal characteristics of its size (or) shape, that would not be obvious from the overhead view alone
- Shadows can provide information about an object's height and shape

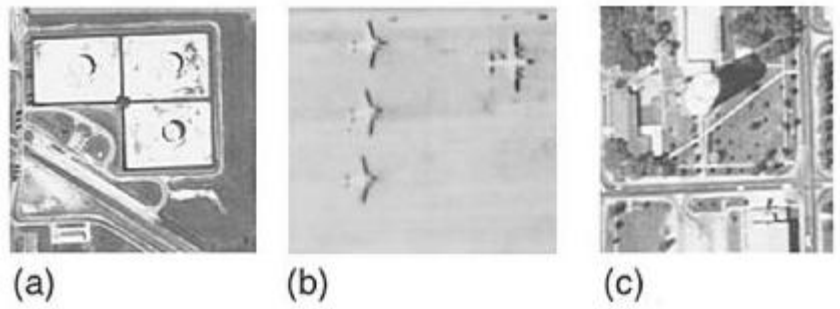


IMAGE INTERPRETATION ELEMENTS

PATTERN

- Arrangement of individual objects into distinctive recurring forms that facilitate their recognition on aerial imagery
- Pattern usually follows a functional relationship between the individual features
- Man-made features, such as cities tend to have very regular patterns, while natural features do not have regular patterns

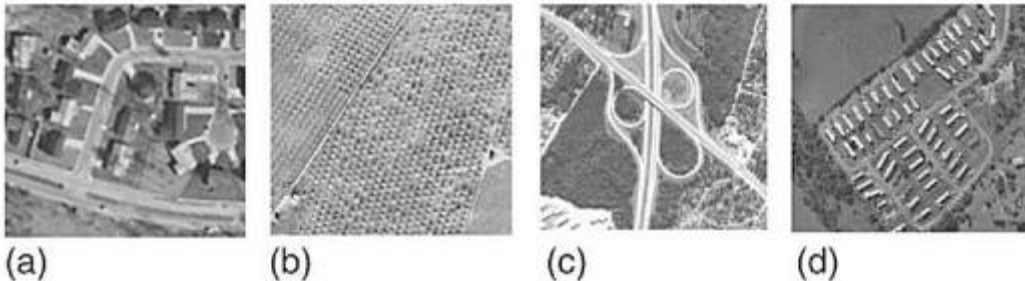


IMAGE INTERPRETATION ELEMENTS

ASSOCIATION

- Specifies the occurrence of certain objects or features, usually without strict spatial arrangement implied by a pattern
- You can identify an object by what is surrounding it, or what it is associated with



IMAGE INTERPRETATION ELEMENTS

SHAPE

- The general outline of objects can help you determine what they are
- Some objects have very distinct shapes while others are more difficult to distinguish
- Man-made features tend to have straight edges while natural features do not



IMAGE INTERPRETATION ELEMENTS

SIZE

- The size of an object can help you interpret what it is
- Size is important in two ways
 - 1) Relative size of an object (or feature) in relation to other objects on the image details image scale and resolution
 - 2) Absolute measurement of size of an object can confirm its identification, permit derivation of quantitative measurements (length, area, ...)



IMAGE INTERPRETATION ELEMENTS

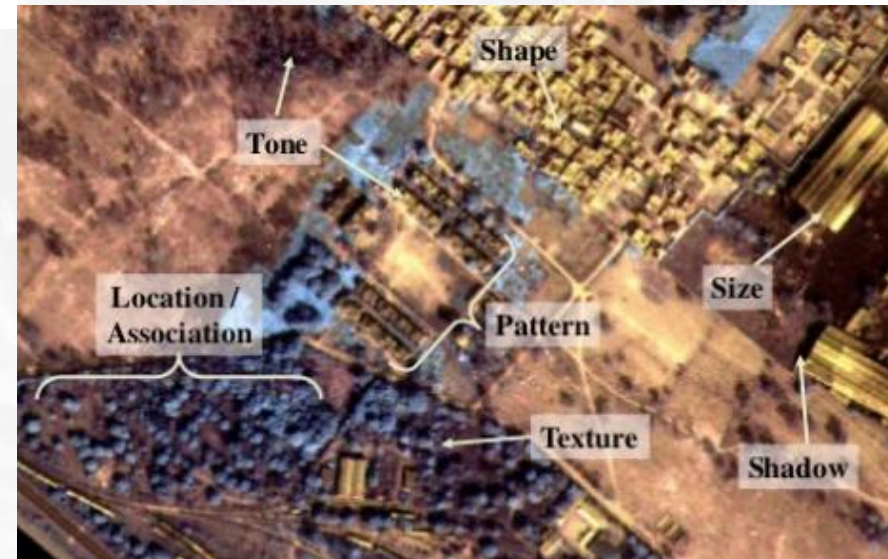
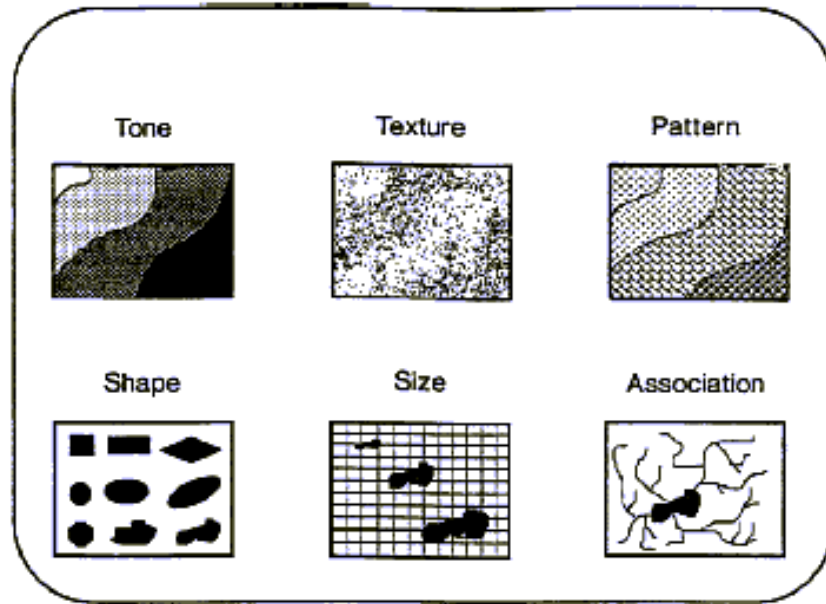


IMAGE INTERPRETATION TASKS

CLASSIFICATION

- Assignment of objects, features, or areas to 'classes' based on their appearance on the image
- Can occur at three levels of confidence
- Detection → Determination of the presence or absence of a feature
- Recognition → Assigning an identity to an object
- Identification → Specify an object (or, feature) with enough confidence and detail to place it in a very specific class

ENUMERATION

- Listing or Counting discrete items visible on an image
- Ability to conduct enumeration depends on ability to accurately identify / classify the objects

IMAGE INTERPRETATION TASKS

MEASUREMENT

- Measurement (Mensuration) has two components

1) Photogrammetric Measurement → Measurement of distances, heights, areas, or volumes of features

Applied to photographs (or) satellite images

2) Photometric Measurement → Measurement of intensity of light in visible region

Densitometers are used to examine the scene brightness

If the measurement is outside the visible spectrum,

radiometry principles are applied



IMAGE INTERPRETATION TASKS

DELINEATION

- The process of outlining (making boundaries) of features as they are observed on remotely sensed images
- Delineation is the process of identifying the edges or boundaries between separate areas
- Problems during delineation
 - 1) Selection of appropriate levels of generalization (when boundaries are intricate)
 - 2) Placement of boundaries when there is a gradation (rather than a sharp edge)

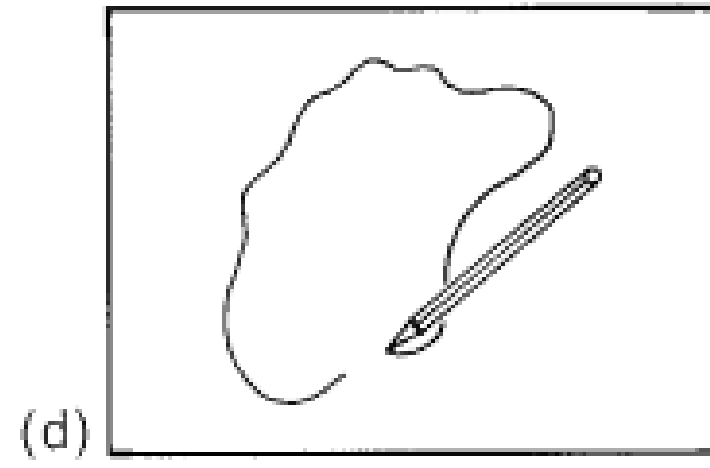
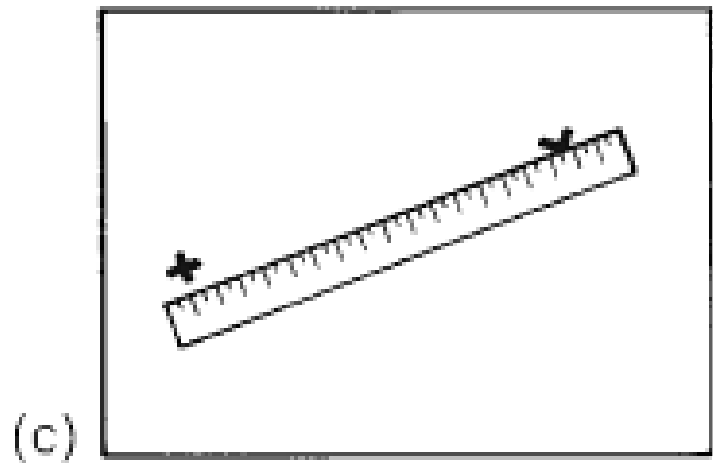
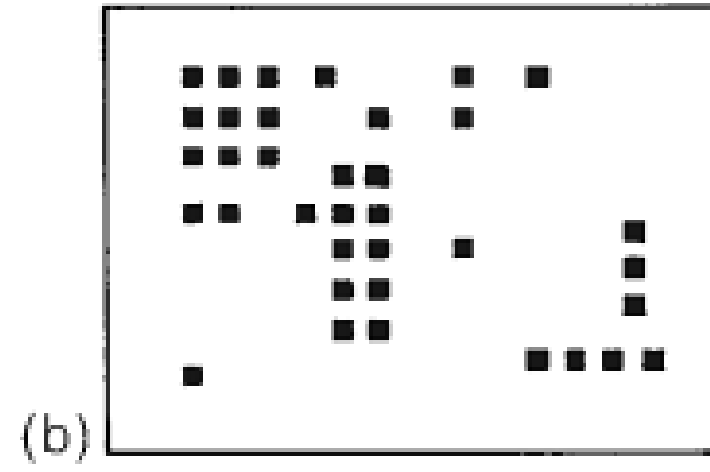
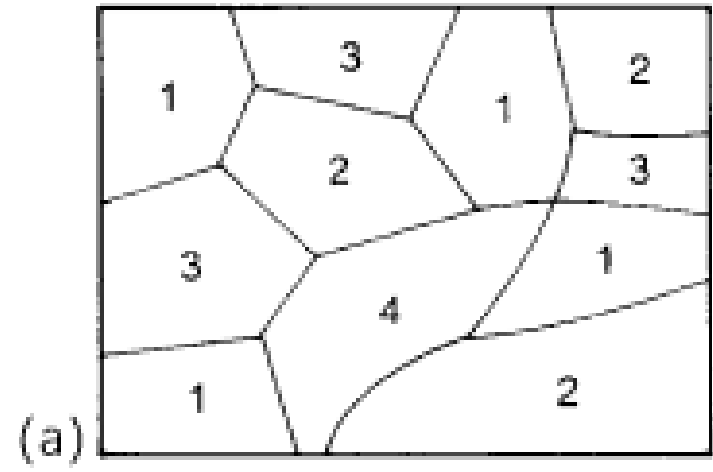


IMAGE INTERPRETATION KEYS

- Reference material designed to permit rapid and accurate objects or features represented on aerial images
- A KEY consists of TWO parts
 - 1) A collection of annotated or captioned images, or stereograms
 - 2) A graphic or word description, possibly including sketches and diagrams
- Image interpretation keys are valuable aids for summarizing complex information portrayed as images
- Image Interpretation Keys serves two purposes
 - 1) Means of training inexperienced personal
 - 2) Reference aid for experienced interpreters

IMAGE INTERPRETATION KEYS

Table 7.5.2 A sample of LANDSAT image's interpretation key

	band 4	band 5	band 6	band 7	457(BGR)	457(RGB)
Snow	PW	PW	PW	PW	PW	PW
Cloud	W	W	W	W	W	W
Haze	W	W	-	-	W	W
Forest	DGR	BL	W	W	R	G
Grass	GR	DG	W	W	P	BY
Bare land	GR	W	W	W	W	W
Wet land	GR	W	GR	DGR	LB	RP
Urban	GR	W	GR	DGR	LB	RP
Water	DGR	BL	BL	BL	B	BP
Shadow	BL	BL	BL	BL	BL	BL

PW: pure white

R: red G: green

LB: light blue

W: white

B: blue

RP: reddish purple

DGR: dark gray

P: pink

BY: brandish yellow

GR: gray

BL: black

IMAGE INTERPRETATION KEYS

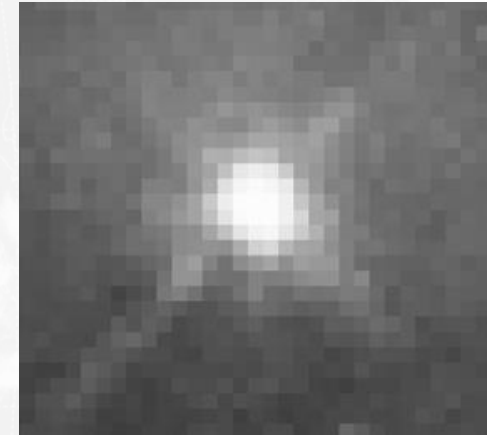
Table 7.5.1 Interpretation keys for forestry

species	crown shape	edge of crown	tone	pattern	texture
ceder	conical with sharp spear	circular and sharp	dark	spotted grain	hard and coarse
cypress	conical with round crown	circular but not sharp	dark but lighter than ceder	spotted	lard and fine
pine	cylindrical with shapeless crown	circular but unclear	light and unclear	irregularly spotted	soft but coarse
larch	conical with unclear crown	circular with unclear edge	lighter than cypress	spotted	soft and fine
fir/spruce	conical with wider crown	circular with zigzag edge	dark and clear	irregular	coarse
deciduous	irregular shapes	unclear	lighter	irregular	coarse

(by country of Japan Association of Forestry)

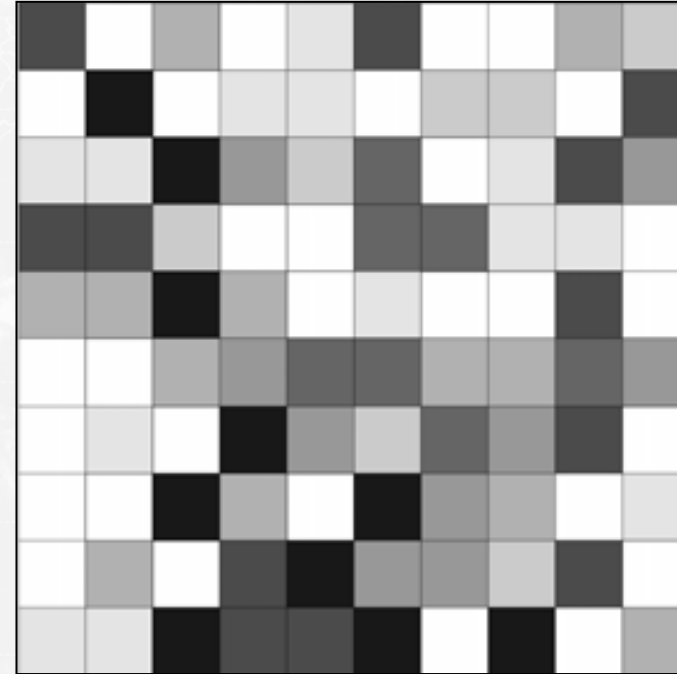
IMAGE ENHANCEMENT

- An image is an array, or a matrix, of square pixels (picture elements) arranged in columns and rows.
- In a (8-bit) image each picture element has an assigned intensity that ranges from 0 to 255.
- A grey scale image (black and white image), will also include many shades of grey.



8-Bit GreyStyle Image

- *Each pixel has a value from 0 (black) to 255 (white).*
- *The possible range of the pixel values depend on the colour depth of the image,*
- *here 8 bit = 256 tones or grey scales.*



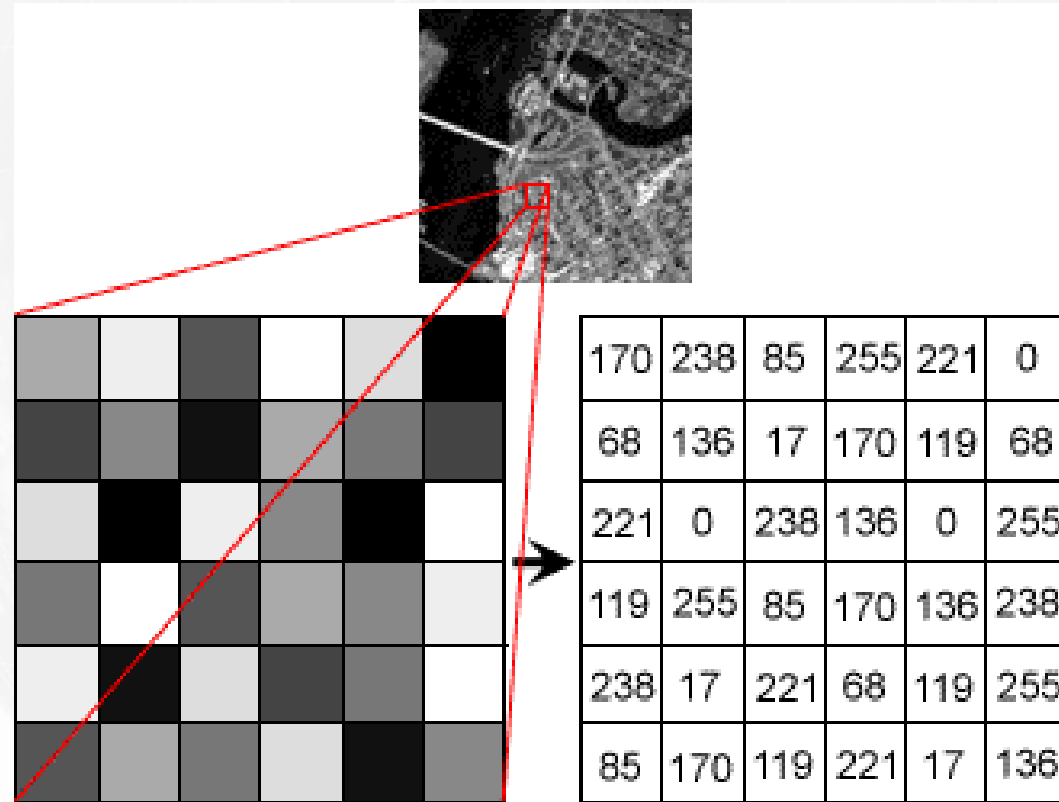
254	107
255	165

PIXEL VALUES

- *Pixel Values*: The magnitude of the electromagnetic energy (or, intensity) captured in a digital image is represented by positive digital numbers.
- The *digital numbers* are in the form of *binary digits* (or '*bits*') which vary from 0 to a selected power of 2

<u>Image Type</u>	<u>Pixel Value</u>	<u>Color Levels</u>
8-bit image	$2^8 = 256$	0-255
16-bit image	$2^{16} = 65536$	0-65535
24-bit image	$2^{24} = 16777216$	0-16777215 16 million colors!!!

DIGITAL NUMBER



TRUE COLOUR IMAGE



- A true-colour image assembled from *three* greyscale images coloured red, green and blue.
- *Such an image may contain up to 16 million different colors.*

IMAGE ENHANCEMENT

- Image Histogram: For every digital image the pixel value represents the magnitude of an observed characteristic (brightness level)
- An image histogram is a *graphical representation* of the brightness values that comprise an image.
- The *brightness values (i.e. 0-255)* are displayed along the *x-axis* of the graph. The *frequency of occurrence* of each of these values in the image is shown on the *y-axis*.



8-bit image
(0 - 255 brightness levels)

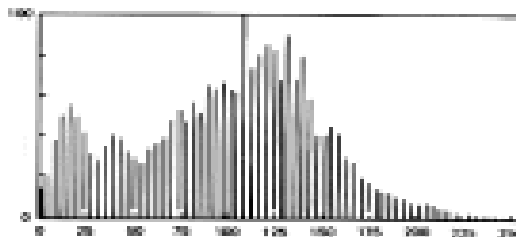


Image Histogram
x-axis = 0 to 255
y-axis = number of pixels

Contrast Stretching

- **Contrast stretching** : Stretching the gray levels in an image to increase the contrast between the features
- Used to increase the dynamic range of gray levels in the image

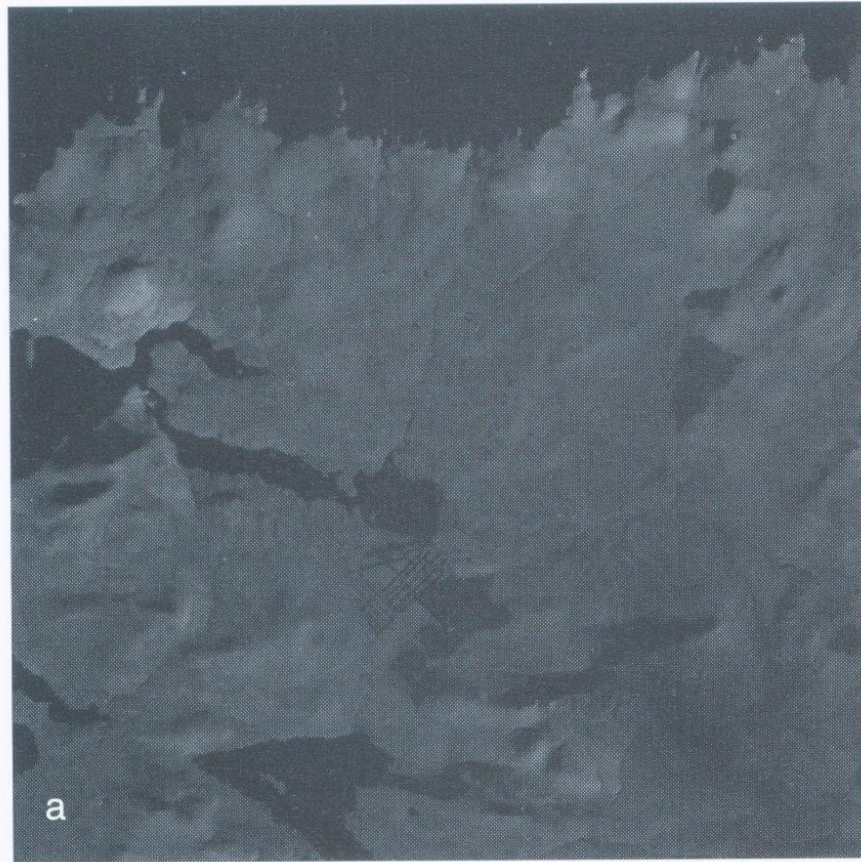
Example:

Number of gray levels possible in an 8-bit system = 256

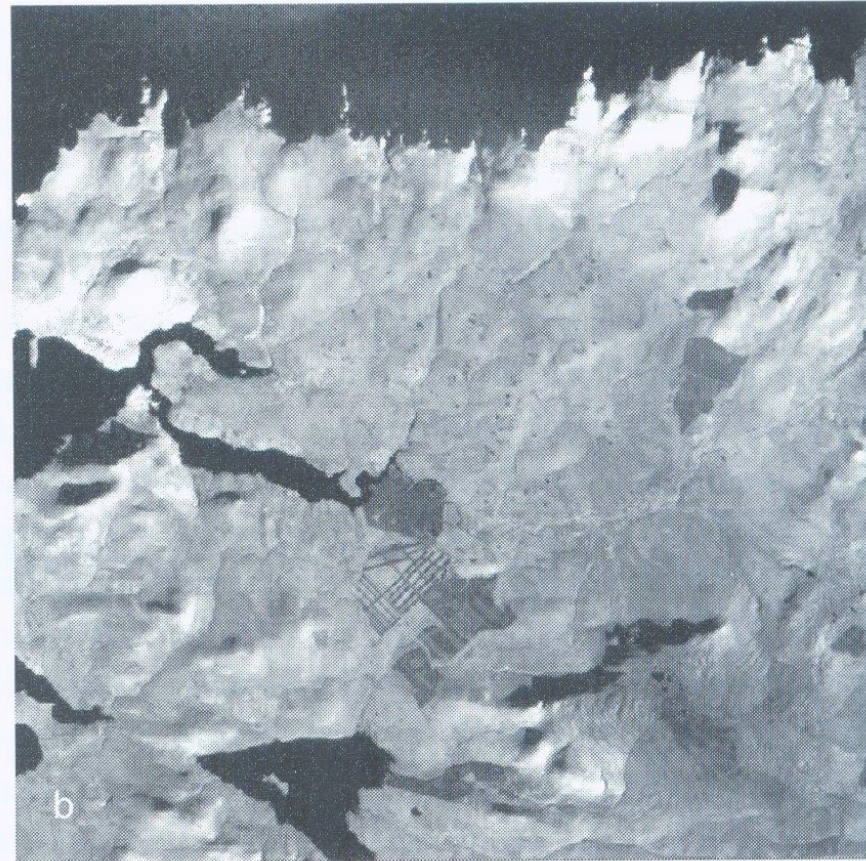
The image pixel values are stretched to use the entire gray level range

- Transformation functions are used to convert the values of the original image into a corresponding value in the output image
 - Linear or Non-linear functions

Contrast Stretching...



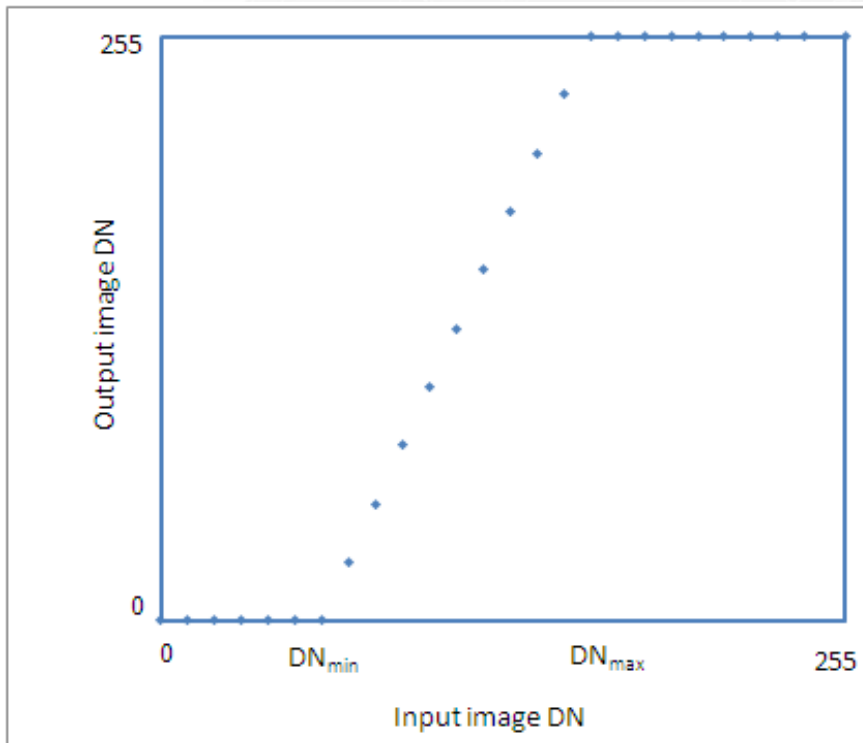
Unstretched Landsat TM Band-5 image



Contrast-stretched image

Linear Contrast Stretching

- Values in the original image are **expanded uniformly to fill the total range** of the output device



DN = Digital Number of the pixel

DN_{st} = DN in the enhanced output image

DN_{max} = Maximum DN in the original image

DN_{min} = Minimum DN in the original image

Linear Contrast Stretching ...

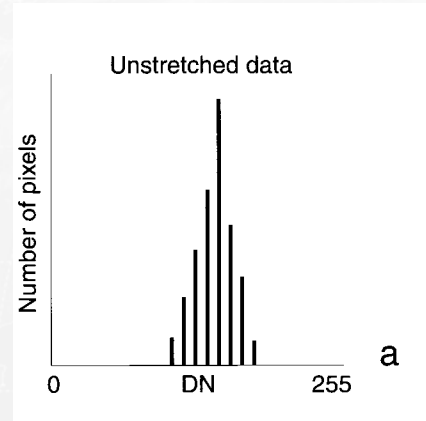
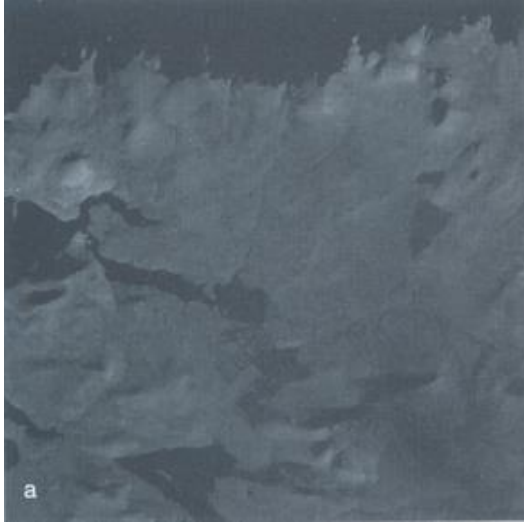
- Linear transformation

$$DN_{st} = 255 \times \frac{(DN - DN_{min})}{(DN_{max} - DN_{min})}$$

Corresponds to the maximum
number of display levels

0-255 in an 8-bit system
0-128 in a 7-bit system

Landsat TM Band-5 image and Histogram



Linear Contrast Stretching...

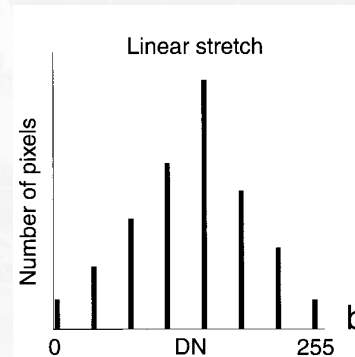
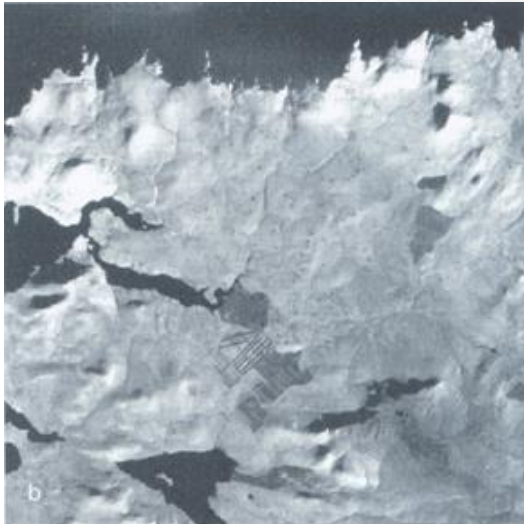
Displayed in an 8-bit system

Image is vague

DN values range from 60-158

0-59 and 159-255 are not utilized

Linearly stretched Landsat TM Band-5 image and Histogram



DN values are stretched to 0-255

Contrast is improved

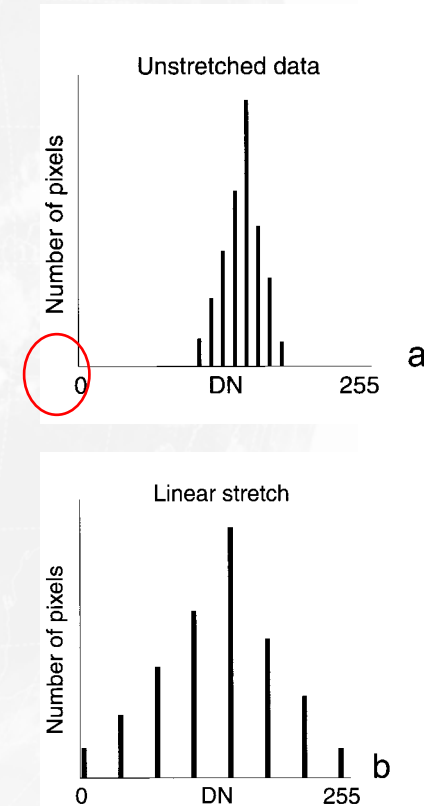
Light tones appear lighter

Dark tones appear darker

Linear Contrast Stretching...

Limitation

- Display levels are not in proportion to the frequency of the DN values
 - Number of pixels having DN values in the range 60-90 are less
 - More number of pixels having higher DN values
 - Linear stretching assigns equal number of display levels to all the ranges
 - Not many display levels are available for the higher DN values

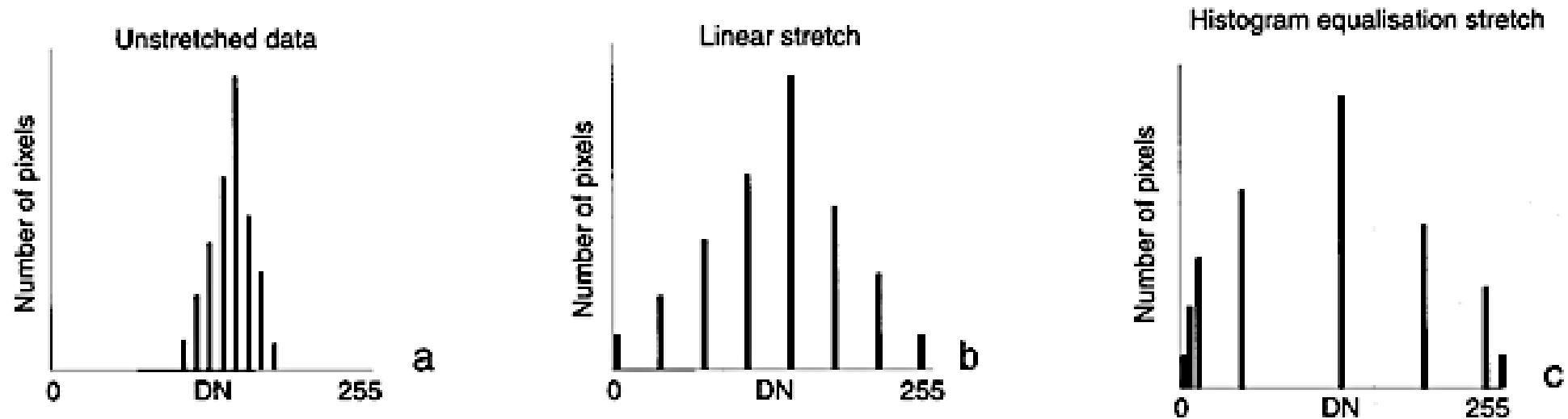


Non-linear Contrast Stretching

- DN values are **not** stretched linearly to uniformly occupy the entire display range
 - Histogram-equalized stretch
 - Piece-wise linear stretch
 - Logarithmic, Power law or Gaussian stretch

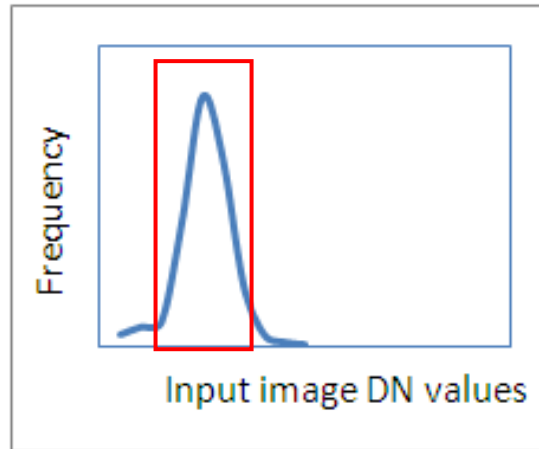
Histogram-Equalized Stretch

- DN values are transformed to have **equal likelihood of occurrence**
- DN values are enhanced based on their frequency in the original image
- DN values corresponding to the peaks of the histogram are assigned to a wider range
- More display levels to the higher frequency region gives better information enhancement



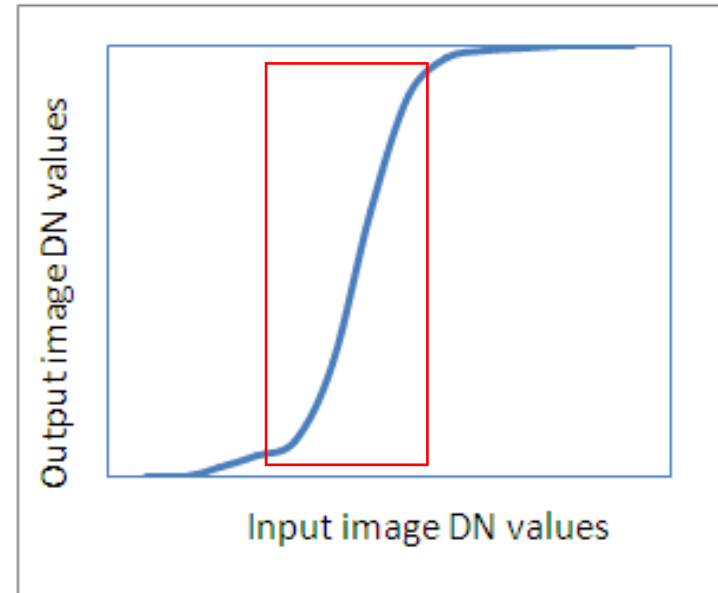
Histogram-Equalized Stretch...

(a)



A sample histogram of an image

(b)



Histogram-equalization stretch

Input DN values corresponding to the peak of the histogram are stretched to a wider range

Histogram-Equalized Stretch...

- DN values are enhanced based on their frequency in the original image

Example: Enhancement in an 8-bit display system

Transformation function

$$DN_{st} = 255 \sum_{j=0}^k \frac{n_j}{N}$$

DN_{st} = Enhanced DN value

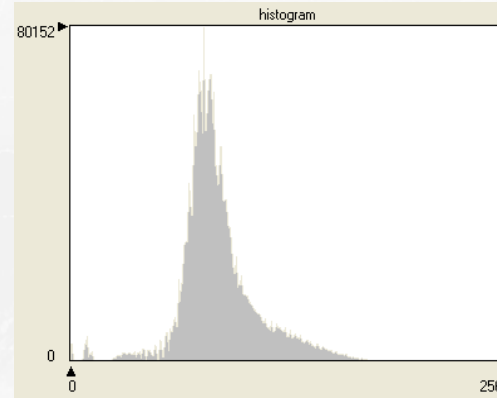
n_j = Number of pixels having DN values in the jth range, in the input image

k = Maximum number of DN ranges in the input image

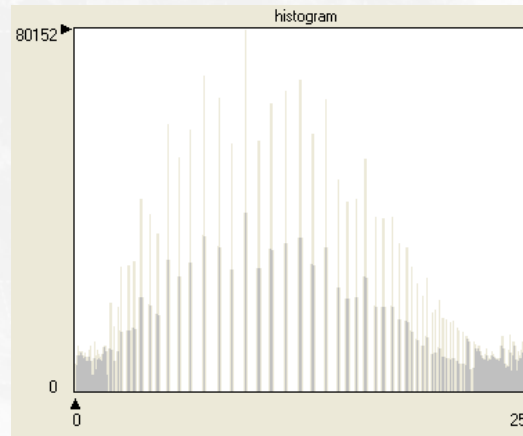
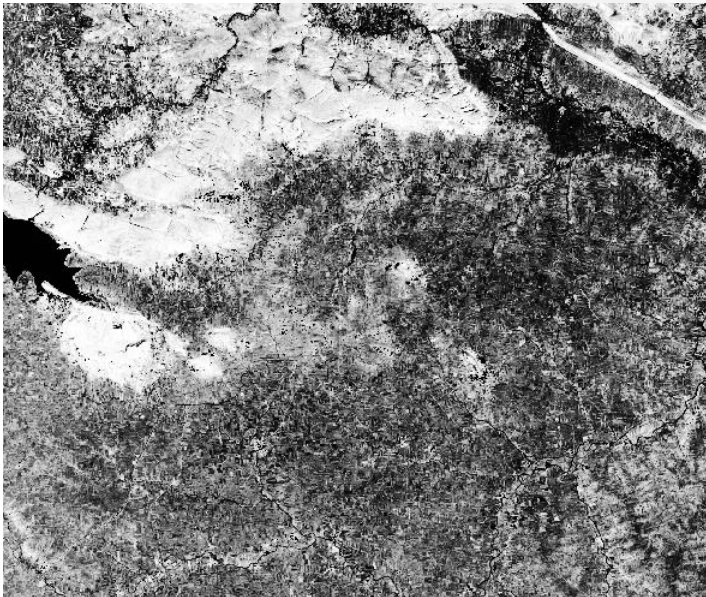
N = Total number of pixels in the input image



Histogram Equalization Stretch...



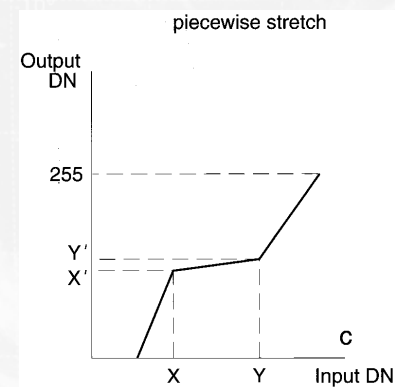
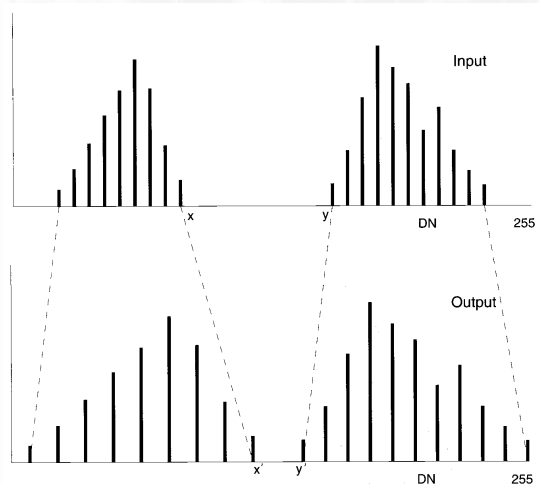
Landsat ETM+band-5 image
&
Histogram



Landsat ETM+band-5 image
after histogram equalization
stretch
&
Histogram

Piece-wise Linear Stretch

- Different linear functions for enhancing the DN numbers in different ranges
- Different parts of the histogram are stretched by different amounts
- Useful in cases where the original image has bi-modal histogram
 - Region between the two modes of the histogram may be compressed
 - Regions corresponding to the histogram peaks may be enhanced

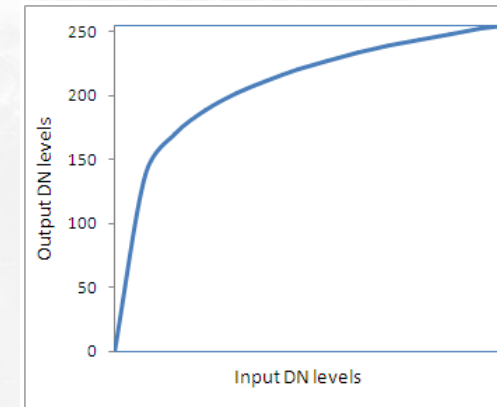


Logarithmic Stretch

- Curves having the shape of the logarithmic function are used
- General form of logarithmic stretching uses the following form

$$DN_{st} = c \log (1+DN) \quad \text{where } c \text{ is a constant}$$

- Smaller values are stretched to a wider range
- A narrower output range is used for higher values.
- Generally used to enhance the information contained in the dark pixels
 - Information contained in the lighter pixels are compressed



Logarithmic Stretch



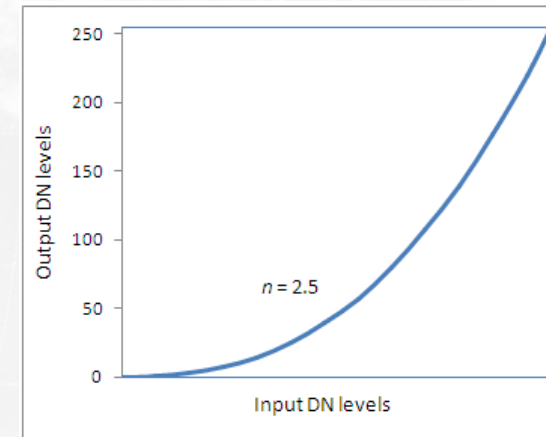
- Smaller values are stretched to a wider range
- A narrower output range is used for higher values.
- Generally used to enhance the information contained in the dark pixels
 - Information contained in the lighter pixels are compressed

Power Law Stretch

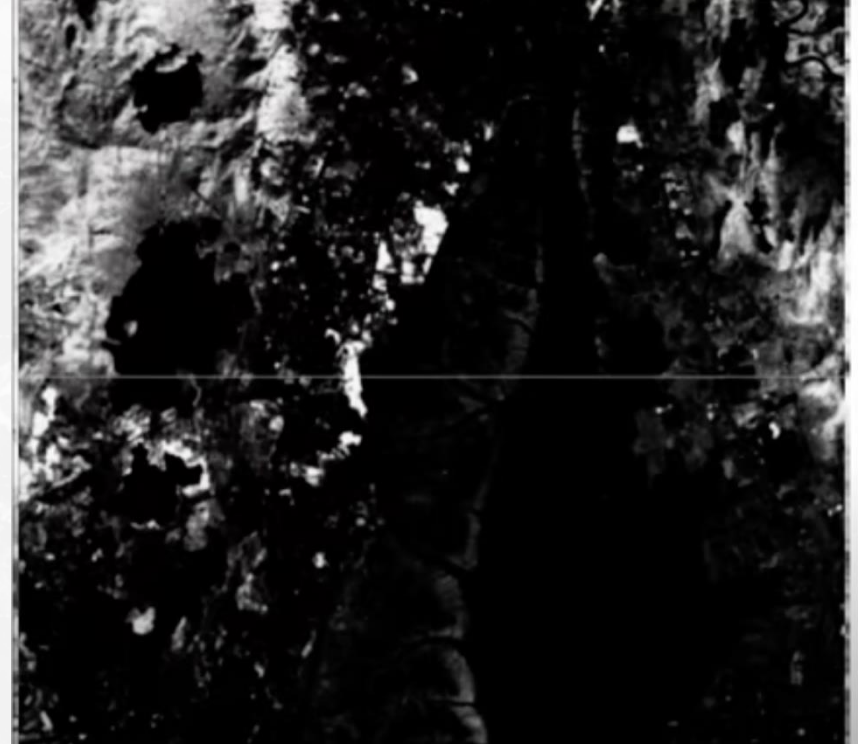
- Curves having the shape of the power function are used
- General form of logarithmic stretching uses the following form

$$DN_{st} = c DN^n \quad \text{where } c \text{ and } n \text{ are constants}$$

- Higher values are stretched to a wider range
- A narrower output range is used for smaller values
- Opposite to logarithmic stretch



Power Law Stretch



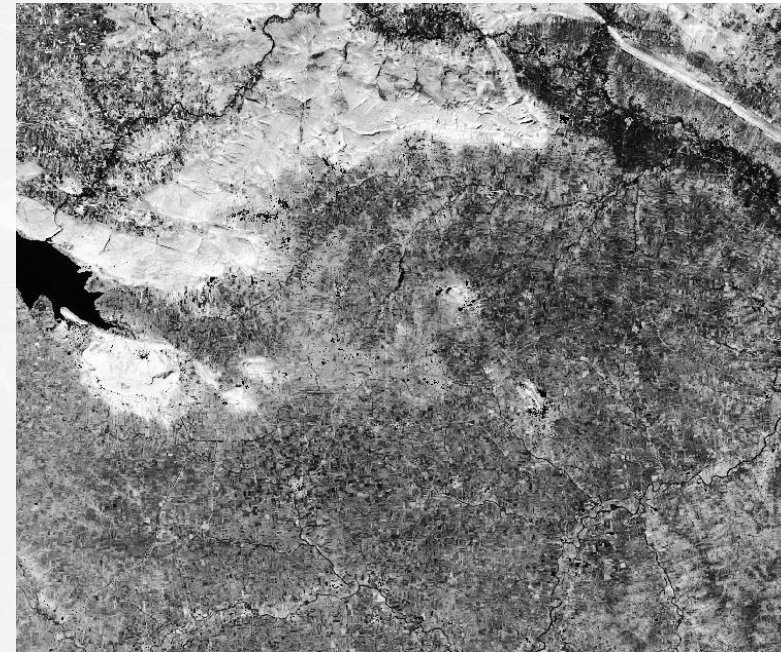
- Higher values are stretched to a wider range
- A narrower output range is used for smaller values
- Opposite to logarithmic stretch

Gaussian Stretch

- Stretched brightness value distribution resembles a normal distribution

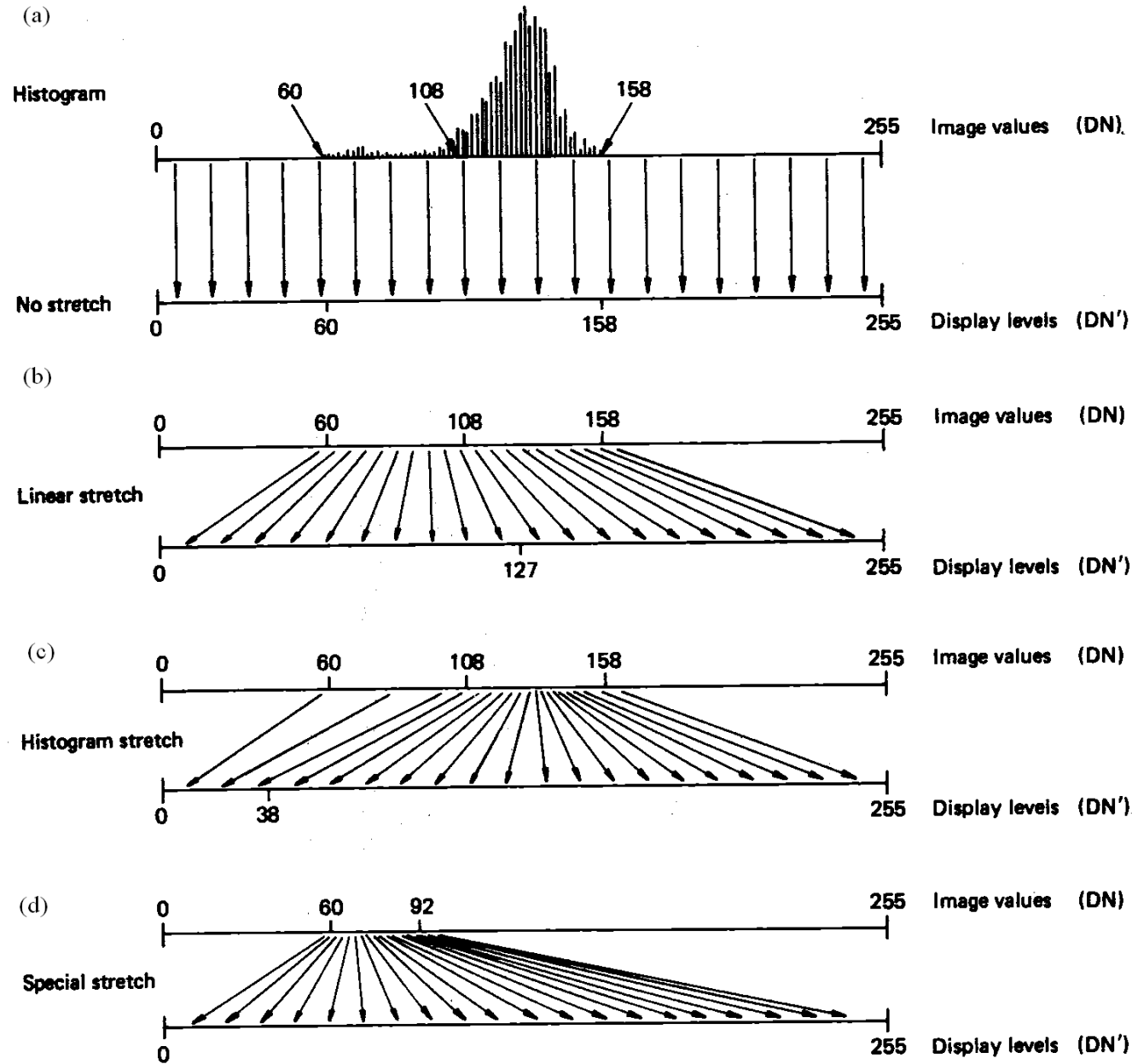


Landsat ETM+band-5 image



Landsat ETM+band-5 image
after applying
Gaussian Stretch

Contrast stretching - Comparison

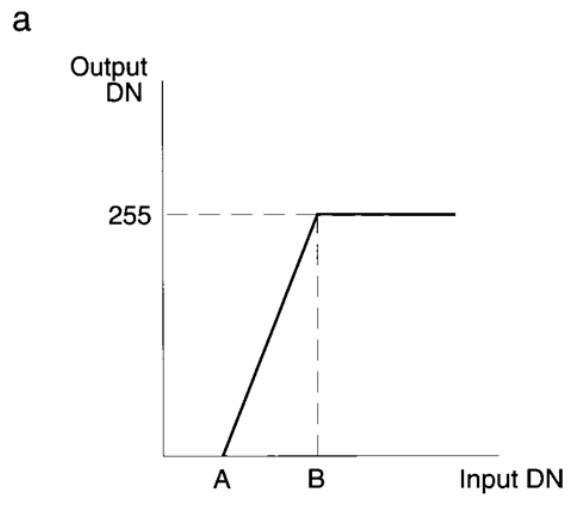


Courtesy:
Lillesand et al., 2004

Look-Up Table

- In contrast stretching, DN values in the original image are transformed into another range
- Well defined transformation functions are used
- Use for transformation function for every pixel of the image takes enormous computation and time
- **Look-up Table (LUT)**
 - Calculate the transformed DN values (DN_{st}) for all possible input DN values in a single stretch
 - Present them in a tabular form, which is the **Look-up Table**
 - For every pixel read the corresponding DN_{st} from the LUT
 - Computations involved and time required are significantly reduced

Look-Up Table...



Transformation function used
for the contrast stretch

b

LOOK-UP TABLE

Input DN	Output DN
20	0
21	3
22	6
23	9
...	
103	249
104	252
105	255
106	255
...	
120	255

LUT corresponding to the
transformation function



Digital Image Processing -Image Enhancement

(iii) Density Slicing, Thresholding, IHS, Time
Composite and Synergic Images

Objectives

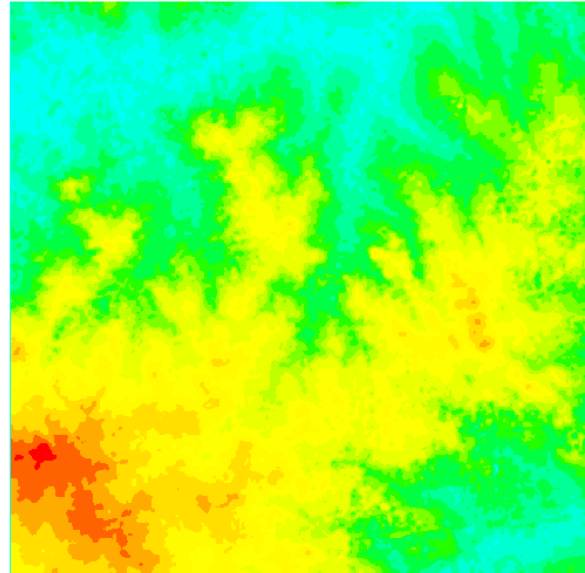
- Image enhancement through color manipulation and the use of additional data
 - ❖ Density slicing
 - ❖ Thresholding
 - ❖ Intensity-Hue-Saturation images
 - ❖ Time composite images
 - ❖ Synergic images

Density Slicing

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- Also known as level slicing
- Pixel values are sliced into different ranges and a single value / color is used for each range in the output image
- Introduces color to a single band image

Density Slicing ...



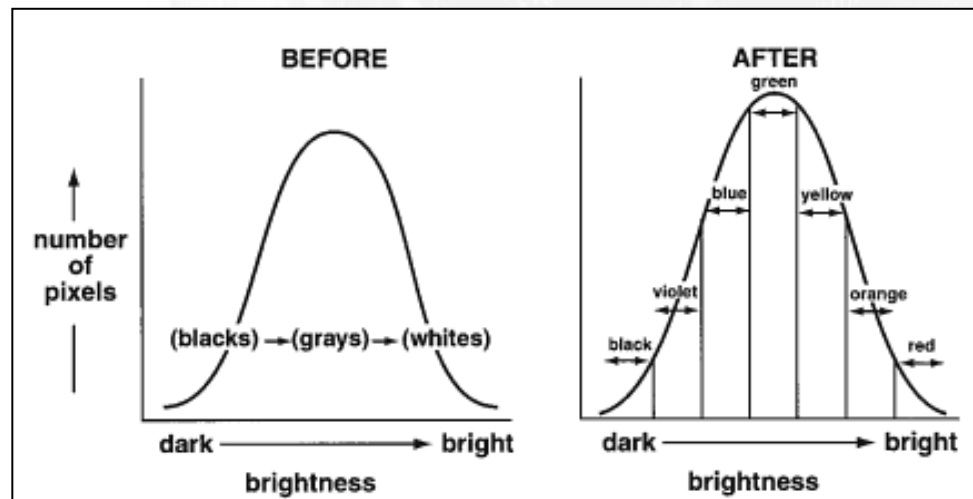
Range	Color
0-620	Blue
620-630	Light Blue
630-640	Green
640-650	Light Green
650-655	Yellow
655-660	Light Yellow
660-665	Orange
665-675	Light Orange
675-685	Red
685-695	Light Red
695-710	Dark Red
710-725	Black
725-750	Dark Blue
>750	White

- 14 levels are used
- Colors are assigned to each level
- Enhances the image information

ASTER GDEM for a small watershed in the Krishna River Basin

- Elevation ranges from 591-770m amsl
- Contrast in the image is not sufficient to clearly identify the variations

Level sliced image



Density Slicing ...

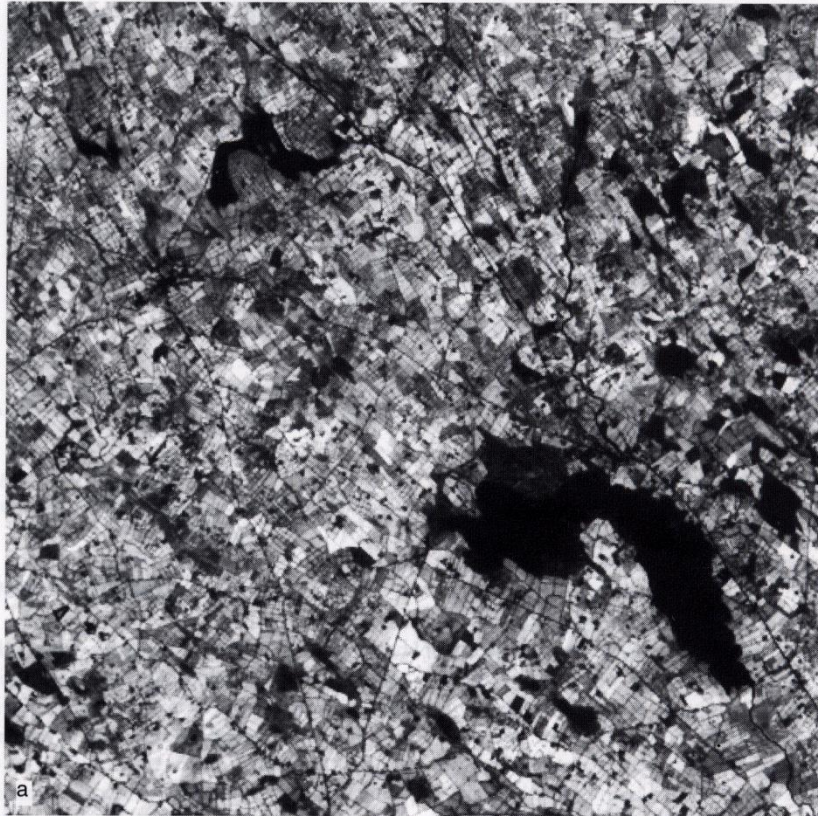
- Advantages
 - Density slicing enhances the image information
 - Useful when the pixel values within a narrow range
 - Enhances the contrast between the levels selected
- Drawback
 - Subtle information loss with in each range or level

Thresholding

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- Divides the image into two classes
 - Pixels having value less than the threshold
 - Pixels having value more than the threshold
- Output image is useful for detailed analysis of each of these classes
 - Required features can be masked out using appropriate threshold

Thresholding...



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Example

Landsat TM Band-5 image

Objective:

Calculate the area of lakes in the image

Thresholding

- Water shows lower DN
- Maximum value for water in the image = 35
- Threshold to mask out water bodies = 35

$DN > 35 \rightarrow 255$ (White)

$DN < 35 \rightarrow 0$ (Black)

Thresholding...

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Thresholding

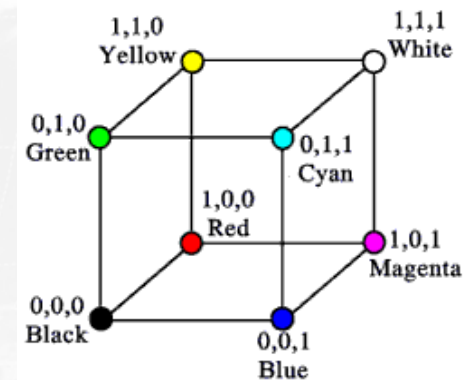
- Threshold to mask out water bodies = 35
 $DN > 35 \rightarrow 255$ (White)
 $DN < 35 \rightarrow 0$ (Black)



Intensity–Hue–Saturation (IHS) Images

RGB Color Composite

- Image is generally the color composite of 3 basic colors: red, blue and green
- Combination of these 3 basic colors at various intensities produces different colors in the image
- For example, each color can vary from 0-255 in an 8-bit display system.
- A color cube, with red, blue and green as its axes, is one way of representing the color composite obtained by adding the three basic colors
(Mentioned in detail in earlier in this module)

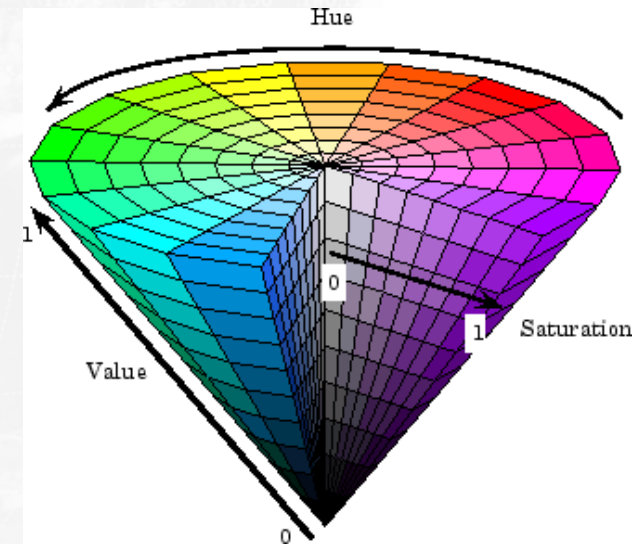


Source: <http://viz.aset.psu.edu>

Intensity–Hue–Saturation (IHS) Images...

IHS Scheme

- An alternate color representation scheme using intensity, hue and saturation
 - Intensity: Brightness of the color
 - Hue: Dominant wavelength contributing to the color
 - Saturation: Purity of the color



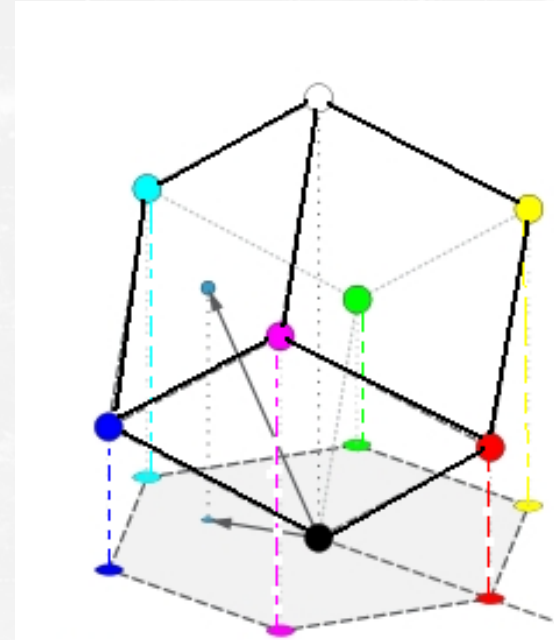
Source: www.mathworks.co.uk

Intensity–Hue–Saturation (IHS) Images...

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Transformation from RGB to IHS color scheme

- RGB color cube is projected into a plane perpendicular to the gray line of the color cube
- Projection gives a hexagon in the plane



Source: :

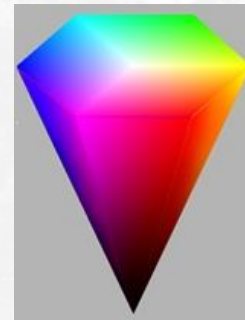
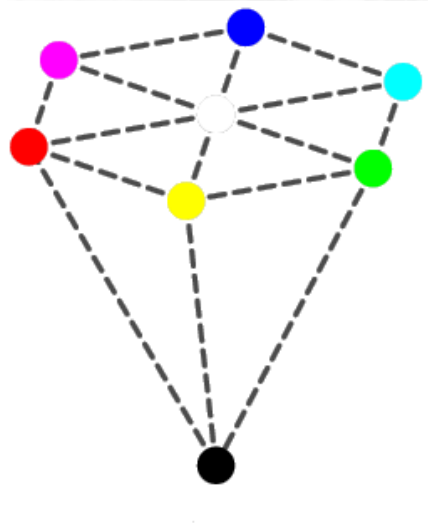
http://en.wikipedia.org/wiki/HSL_and_HSV

Intensity–Hue–Saturation (IHS) Images...

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Transformation from RGB to IHS color scheme

- Size of the hexagon varies with the location of the plane
- Size increases as the plane is moved from black to white
- Such a series of hexagons are combined to form hexacone



Source:

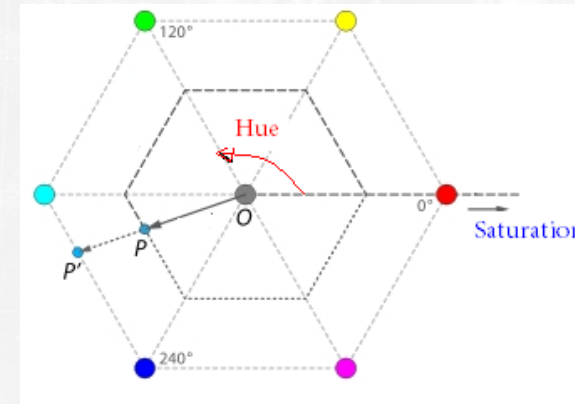
http://en.wikipedia.org/wiki/HSL_and_HSV)

Intensity–Hue–Saturation (IHS) Images...

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Transformation from RGB to IHS color scheme

- Size of the hexagon at any point along the cone is determined by the intensity
- Within each hexagon, Hue increases counterclockwise from the axis corresponding to red
- Within each hexagon, Saturation is the length of the vector from the origin.



Source:

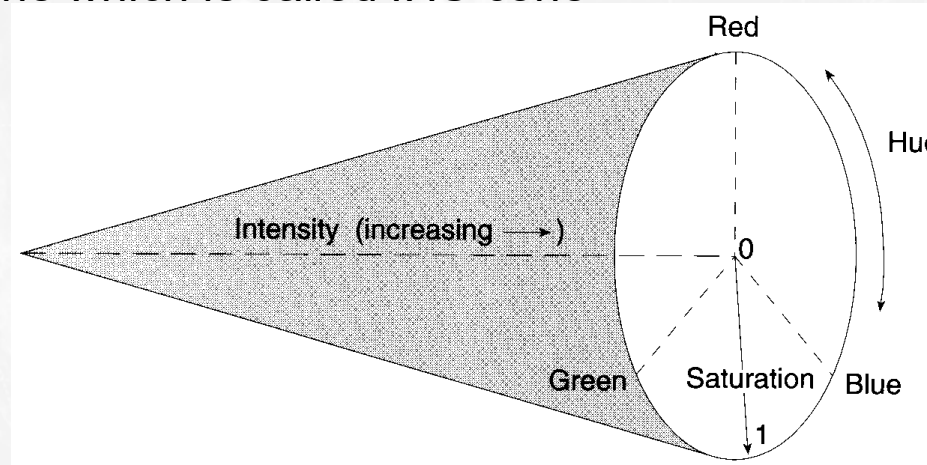
http://en.wikipedia.org/wiki/HSL_and_HSV)

Intensity–Hue–Saturation (IHS) Images...

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Transformation from RGB to IHS color scheme

- Circular planes are also used to represent the IHS transformations
- This gives a cone which is called IHS cone

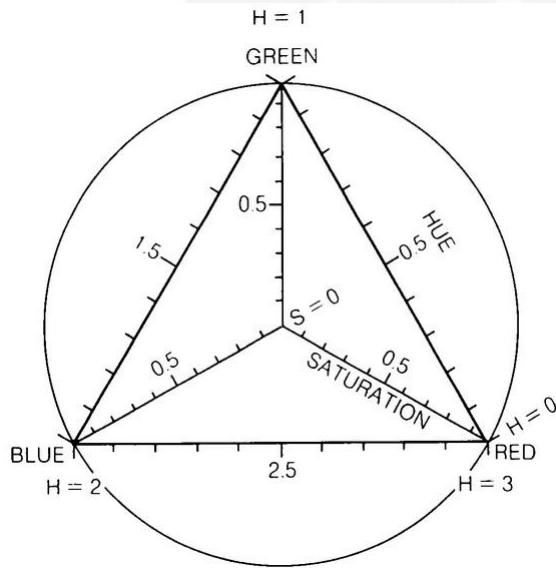


Intensity–Hue–Saturation (IHS) Images...

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Transformation from RGB to IHS color scheme

- The relationship between the RGB components and the corresponding IHS components is represented using a triangle



$$I = (R + G + B) / 3$$

$$H = \begin{cases} \theta & \text{if } B \leq G \\ 360 - \theta & \text{if } B > G \end{cases}$$

$$S = \frac{I - [\min(R, G, B)]}{I}$$

$$\theta = \cos^{-1} \left\{ \frac{\frac{1}{2} [(R - G) + (R - B)]}{[(R - G)^2 + (R - B)(G - B)]^{1/2}} \right\}$$

Intensity–Hue–Saturation (IHS) Images...

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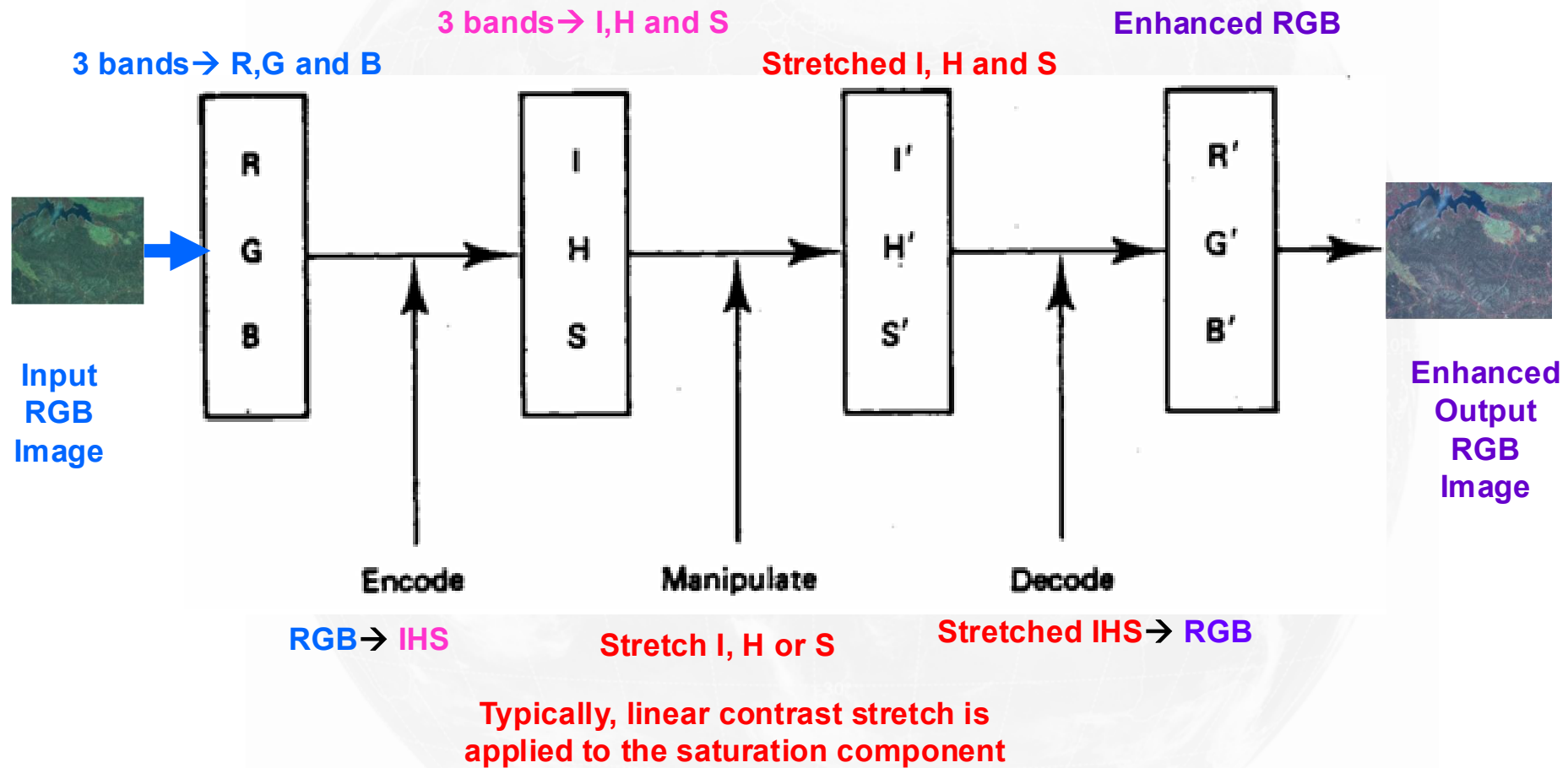
Problem associated with the RGB display scheme

- Typical MSS data displayed using the RGB scheme lacks saturation
 - Due to the high degree of correlation between the spectral bands
 - Example: High reflectance in Blue is combined with high reflectance in Green and Red as well, reducing the purity of the color

Image enhancement through IHS transformation

- RGB scheme may be converted into IHS scheme
- Saturation component can be stretched separately, without affecting Intensity and Hue
- Enhances color in the output

Intensity–Hue–Saturation (IHS) Images...

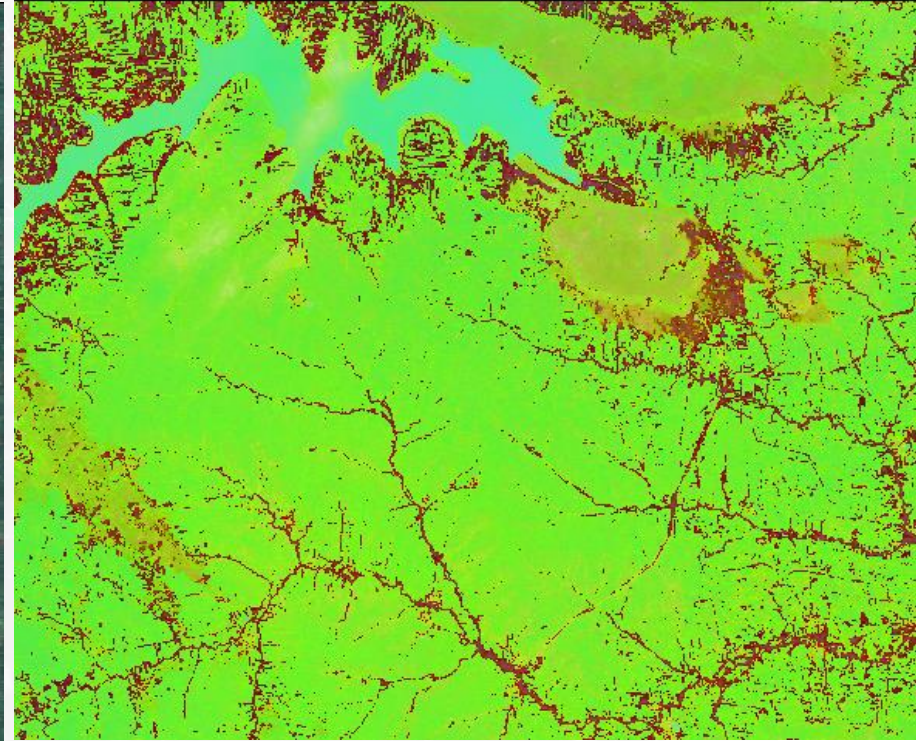


Intensity–Hue–Saturation (IHS) Images...

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Landsat FCC image



IHS image

The saturation component is linearly stretched

Intensity–Hue–Saturation (IHS) Images...

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Landsat FCC image before
enhancement



Landsat FCC image
After color enhancement through
IHS transformation

Intensity–Hue–Saturation (IHS) Images...

Advantages of enhancement through IHS transformation

- IHS transfer gives more control over the color enhancement
- Transformation from RGB scheme to IHS scheme gives the flexibility to vary each component of the IHS system separately without effecting the other
- The IHS transformed image can be used to generate synergic images

Synergic Images

- Images generated by combining information from different sources
- Images of different spatial and spectral resolutions are merged to enhance the information contained in an image
- FCC can be produced by considering any three bands (may be of different spectral or spatial resolution)
- Generally IHS scheme is used



Example -1: Clouds

Synergic Images

PAN Sharpening

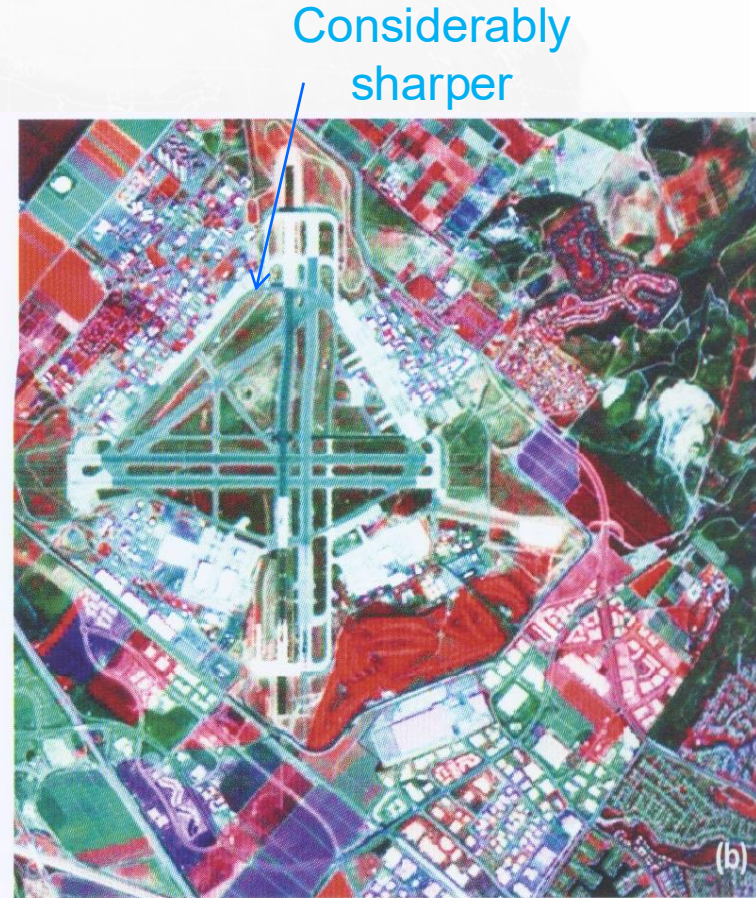


Image source: QuickBird, (C) DigitalGlobe

Synergic Images ...



Landsat TM image



TM/SPOT fused data for an airport
southeast of Los Angeles

Synergic Images ...

Example : Merge PAN data with LISS III MSS data

- Resolutions

- IRS 1C PAN

- 5.8m spatial resolution
 - poor spectral resolution

- IRS 1C LISS-III

- 23m spatial resolution
 - better spectral resolution

- Intensity component is substituted from the LISS-III image
- Higher spatial resolution from the PAN data is combined with better spectral resolution from the LISS-III data

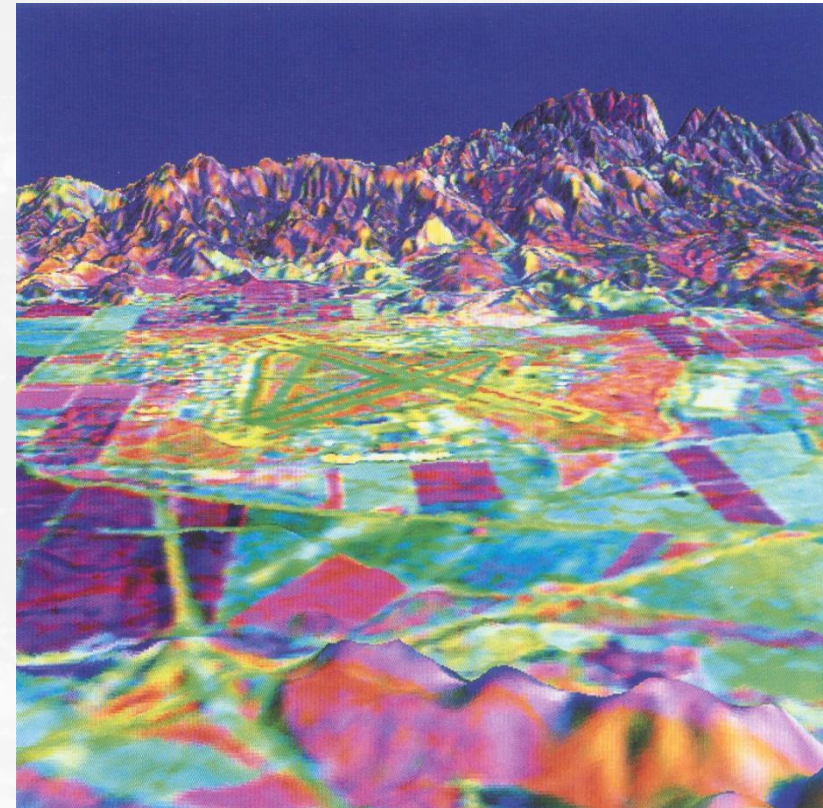


IRS-1C(LISS III and PAN) digitally merged and enhanced image of Hyderabad city

Synergic Images ...

- Non-remote sensing data are also be merged
 - Example Satellite images merged with DEM

Perspective view :
Southeast of Los Angeles produced by
draping TM and radar data over a digital
elevation model and viewing from the
southwest



Time Composite Images

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- Images generated by integrating information from images obtained at multiple time periods
- Missing data in one images is filled using images obtained on other dates

Example:

- NOAA AVHRR data with cloud cover- cloud cover in a small portion of an image
- Obtain data for the cloud covered area from the successive images



Example -1: Clouds

Time Composite Images

- Steps followed for generating time composite images
 - **Co-register images** acquired over number of days (say 15 days)
 - Area with cloud cover is identified from the first imagery and is replaced by the next imagery of the same area.
 - Cloud cover (if any) from this composite imagery is replaced with the third imagery.
 - This procedure is repeated 15 times (say over 15 days imageries)

NRSC uses such time composited imageries of NOAA AVHRR over 15 days for Agricultural drought assessment and analysis



Digital Image Processing -Image Enhancement

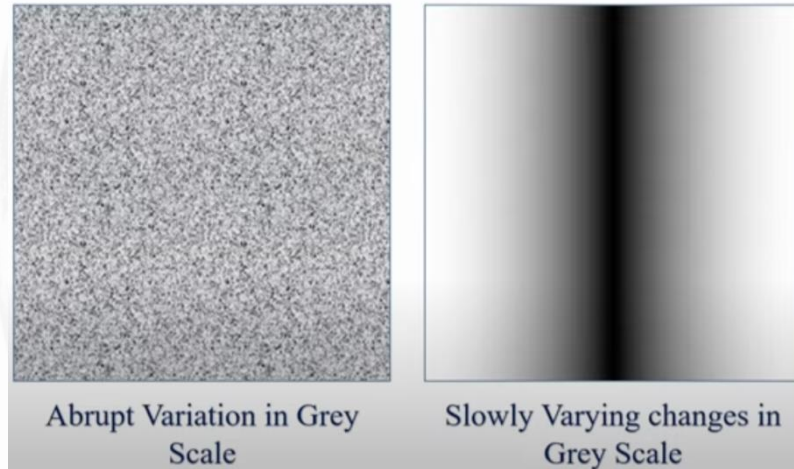
(iv) Filtering and Edge Enhancement

Objectives

- Filtering techniques in image enhancement
 - Convolution filter
- Edge enhancement
 - Directional filter
 - Non-directional filter

Introduction

- Image enhancement: Manipulations to enhance the information in the image
- **Spatial feature manipulation**
 - Processes to **emphasize or deemphasize** data at various spatial frequencies
 - Spatial frequency: Tonal variation in the image



- **Local operations** to manipulate pixel values with respect to the values of the neighboring pixels
- Spatial or frequency domain manipulations
 - **Spatial domain manipulation: Direct manipulation of the pixel values**
 - Frequency domain manipulations: Manipulations on the Fourier transformation of the pixel values

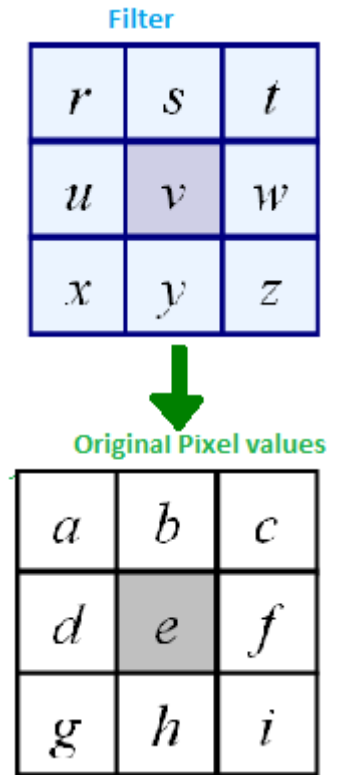
Introduction...

- **Spatial feature manipulation**
 - Filtering techniques
 - Edge enhancement

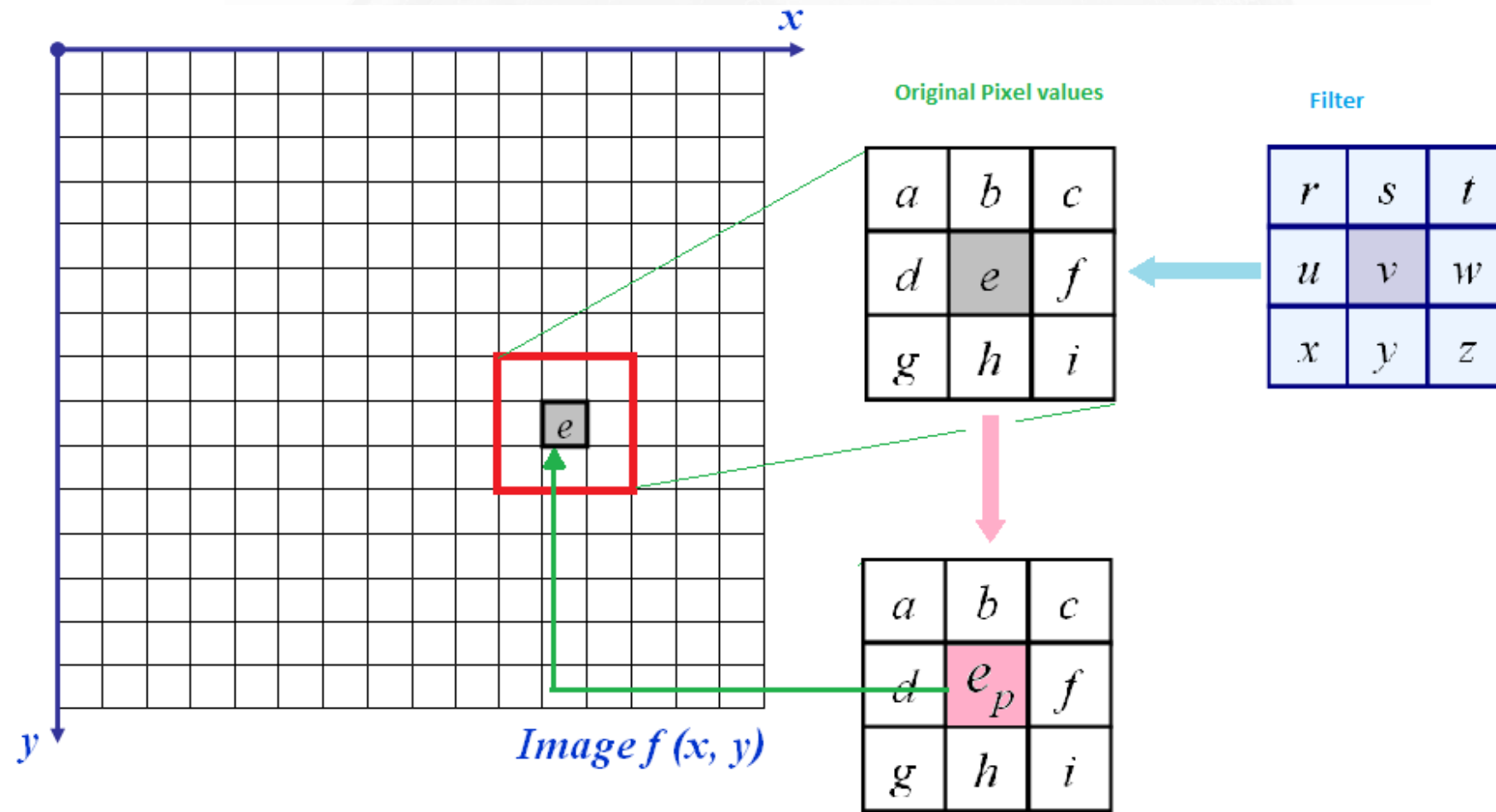
Spatial feature manipulations in the spatial domain is covered in this lecture

Filtering Techniques

- **Filtering** : Selectively enhances or suppresses particular wavelengths or pixel DN
 - Convolution filtering in the spatial domain
 - Fourier analysis in the frequency domain
- **Filter**
 - A regular array or matrix of numbers
 - Values in the filter are called filter coefficients
 - Allows the formation of a new image through simple arithmetic operations using the filter coefficients



Filtering Techniques...



Convoluting

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

Input image

*

1	0	1
0	1	0
1	0	1

Filter



1 _{x1}	1 _{x0}	1 _{x1}	0	0
0 _{x0}	1 _{x1}	1 _{x0}	1	0
0 _{x1}	0 _{x0}	1 _{x1}	1	1
0	0	1	1	0
0	1	1	0	0

Image

4		

Convolved Feature

Convoluting

1 _{x1}	1 _{x0}	1 _{x1}	0	0
0 _{x0}	1 _{x1}	1 _{x0}	1	0
0 _{x1}	0 _{x0}	1 _{x1}	1	1
0	0	1	1	0
0	1	1	0	0

Image

4		

Convolved
Feature

Filtering Techniques ...

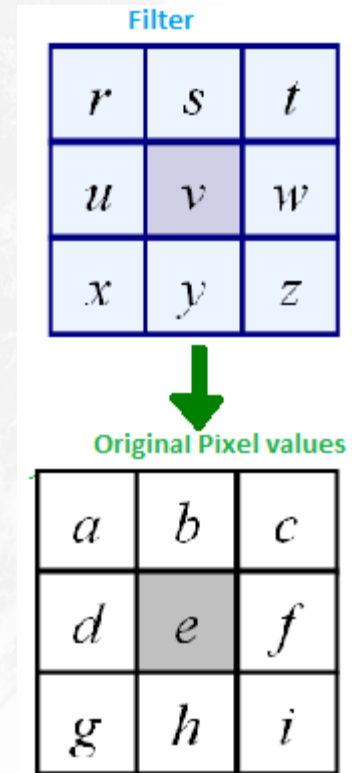
- **Mechanics of spatial filtering**
 - Filter mask is moved over the image
 - For each location of the filter, the value at the centre of the filter mask is modified
 - Pre-defined relationship between filter coefficients and the pixel values is used
- Example : Convolution filter
 - Spatial domain: Manipulates the original pixel values
 - Frequency domain: Manipulates the image obtained by the Fourier transformation of the original image.

Convolution Filter

- Most commonly used for image enhancement
- Filter mask is called convolution mask or convolution kernel
 - Square in shape (3x3, 5x5, 7x7 etc.)
- Linear transformation function is used to derive new pixel value
- Pixel values are multiplied with the kernel coefficients and are averaged

$$e_p = (e.v + a.r + b.s + c.t + d.u + f.w + g.x + h.y + i.z)/9$$

- The kernel coefficients are modified to achieve image enhancement for different purposes



Convolution Filter ...

- Different types of filters
 - High pass filter
 - Low pass filter
 - Directional gradient filter
 - Emboss filter
 - Prewitt gradient filter
 - Sobel filter
 - Laplacian filter

High and Low Pass Filters

- Low pass filter

- Averaging filter: Filter output is the average of all the pixels in the mask
- Reduces the effect of noise component
- Smooth or blurred image, reduced gray level range
- Enhances the large-area brightness regime

1	1	1
1	1	1
1	1	1

Low pass

- High pass filter

- Enhances the higher values
- Deemphasize the lower values

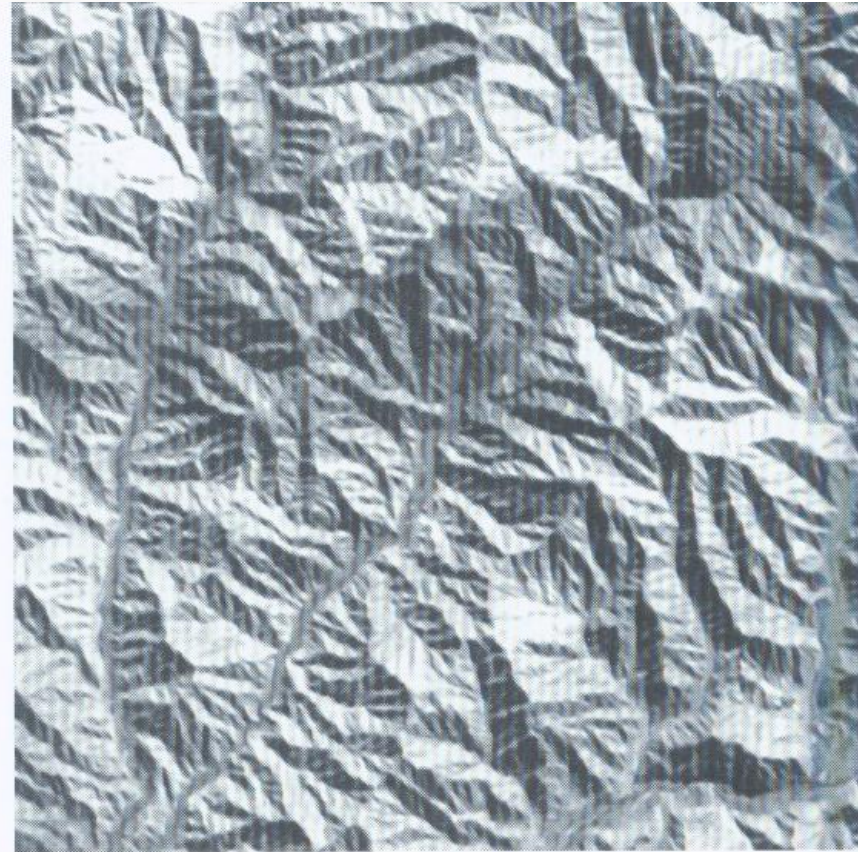
-1	-1	-1
-1	8	-1
-1	-1	-1

High pass

High and Low Pass Filters...

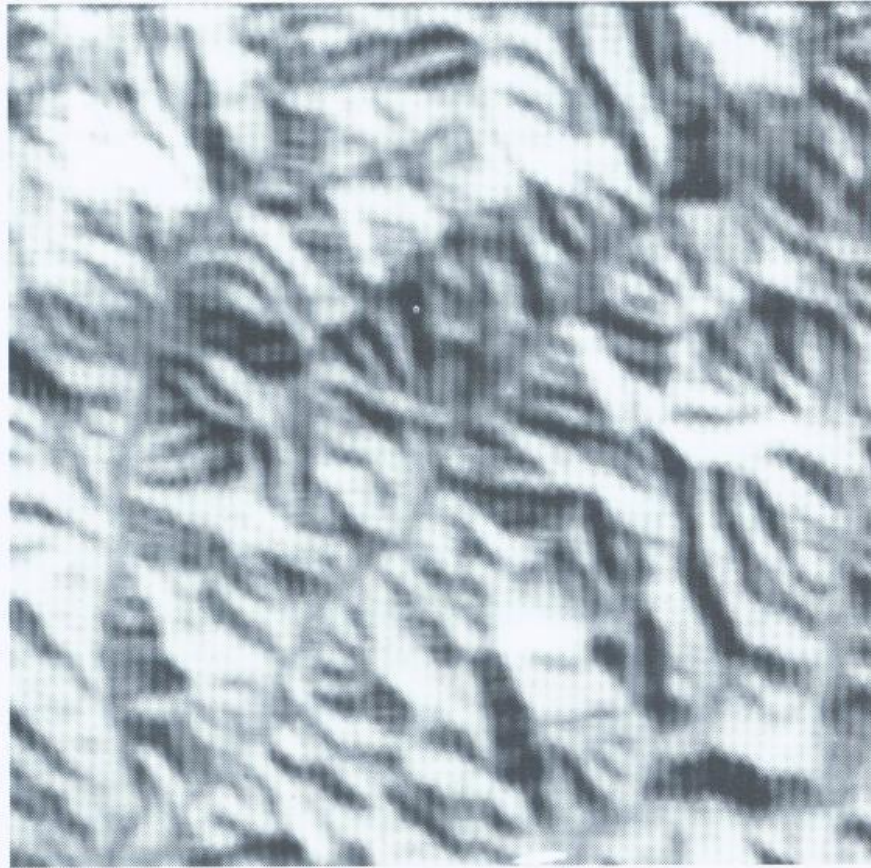


Original image

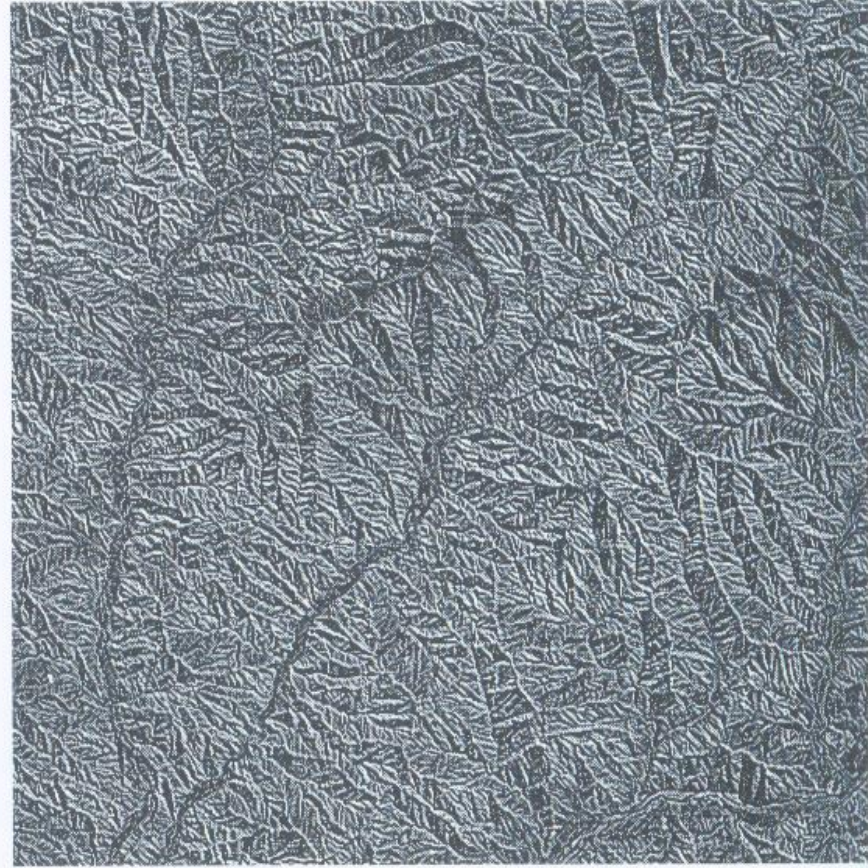


3x3 low pass filtered image

High and Low Pass Filters...

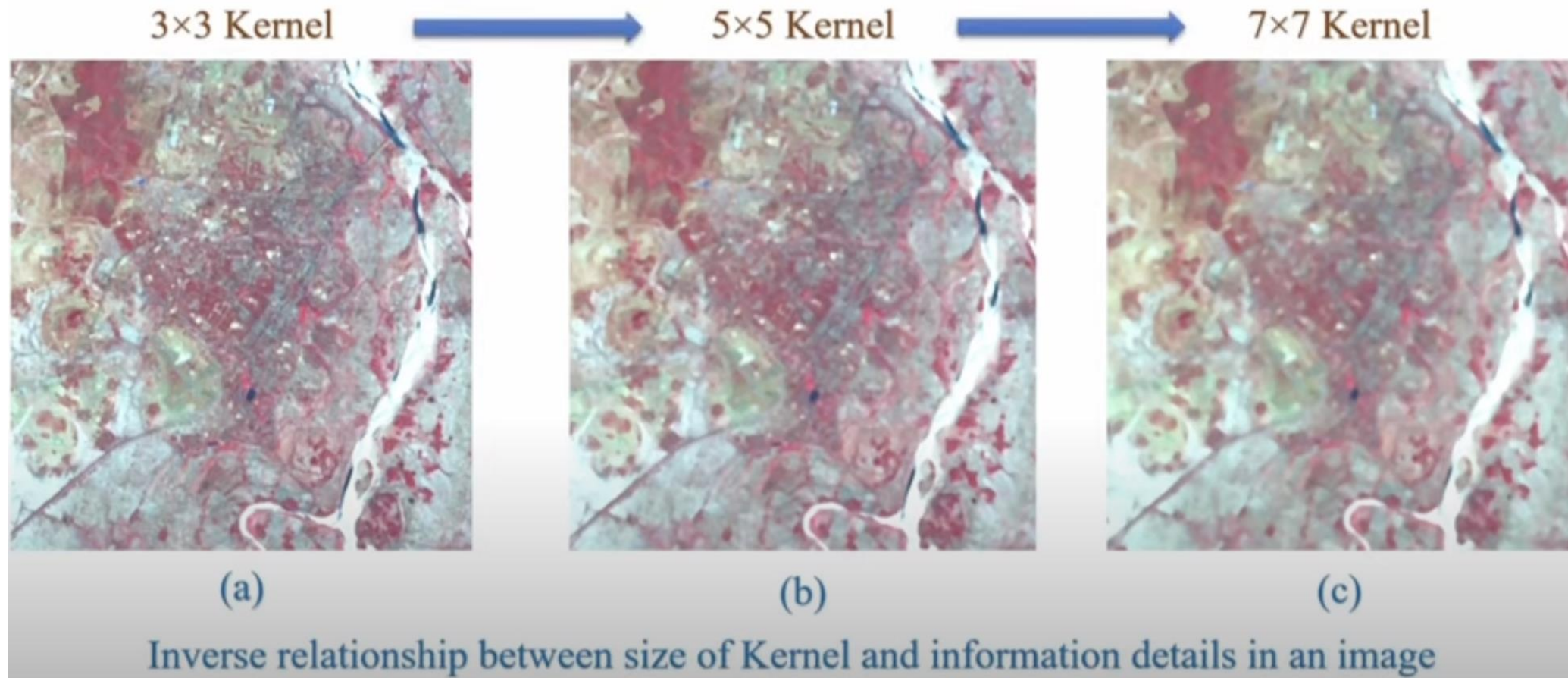


11x11 low pass filtered image



3x3 high pass filtered image

Low Pass Filters...



Directional Gradient and Emboss Filter

- Highlight features in any specific direction

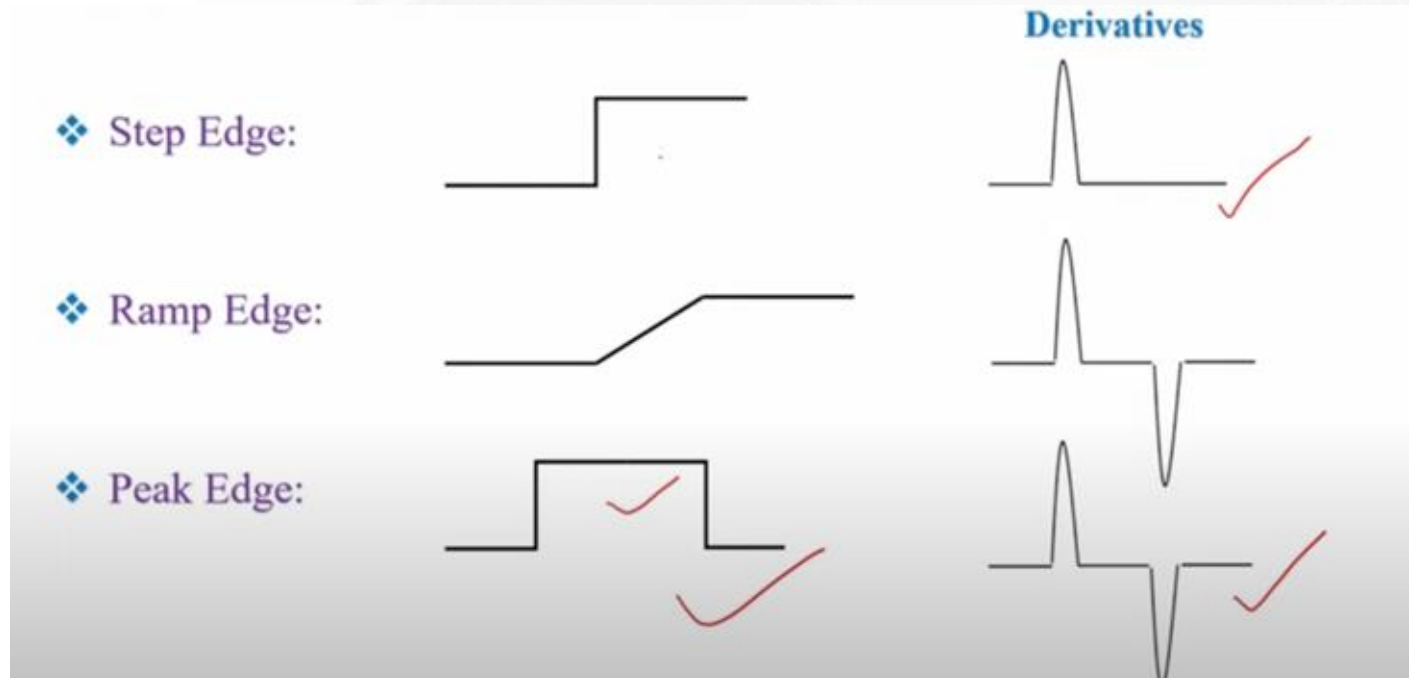
1	0	-1
2	0	-2
1	0	-1

Directional gradient
filter - N/S features
highlighted

0	0	1
0	0	0
-1	0	0

Emboss NW

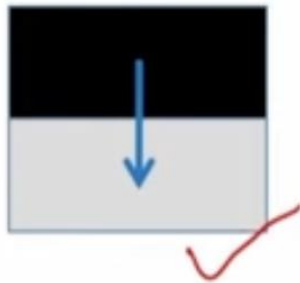
Directional Gradient and Emboss Filter



Directional Gradient and Emboss Filter

Estimates the intensity gradient in two perpendicular directions...

Gradient Directions



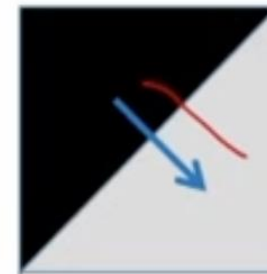
Vertical Gradient



Horizontal Gradient



Diagonal Gradient



- ✓ Two mutually perpendicular gradient detectors are required to detect edges...
- ✓ An edge can be resolved in terms of two orthogonal components...

Directional Gradient and Emboss Filter

- Highlight features in any specific direction

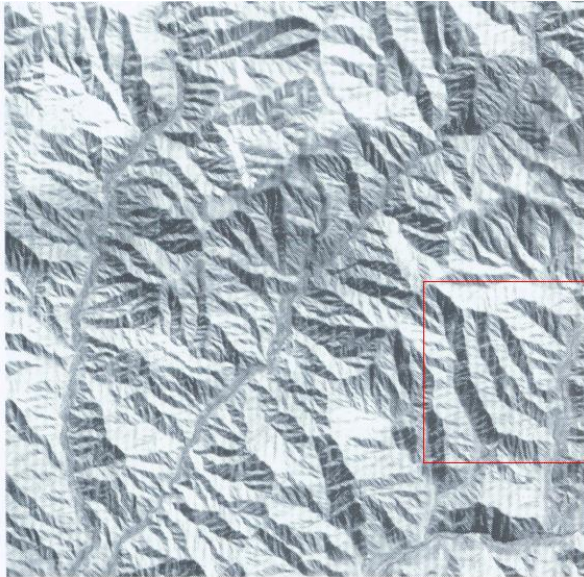
1	0	-1
2	0	-2
1	0	-1

Directional gradient
filter - N/S features
highlighted

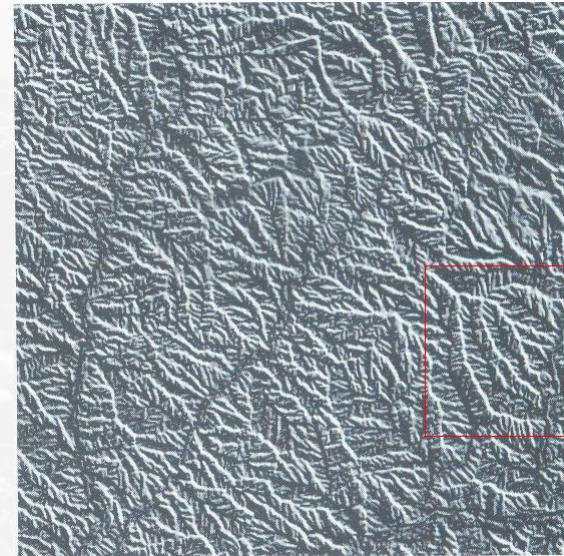
0	0	1
0	0	0
-1	0	0

Emboss NW

Directional Gradient and Emboss Filter...



Original image



Gradients in
the NW
direction are
highlighted

NW emboss filtered image

Prewitt gradient, Sobel gradient, Laplacian Filter

- Estimates the digital gradient in an image
- Prewitt gradient and Sobel gradient use a set of two kernels
- Identify the sharp changes in the pixel values and hence the lineaments
- Useful in edge enhancement and edge detection

0	-1	0
-1	5	-1
0	-1	0

Laplacian filter

-1	0	1
-1	0	1
-1	0	1

x

1	1	1
0	0	0
-1	-1	-1

Prewitt

y

-1	0	1
-2	0	2
-1	0	1

1	2	1
0	0	0
-1	-2	-1

Sobel

Prewitt gradient

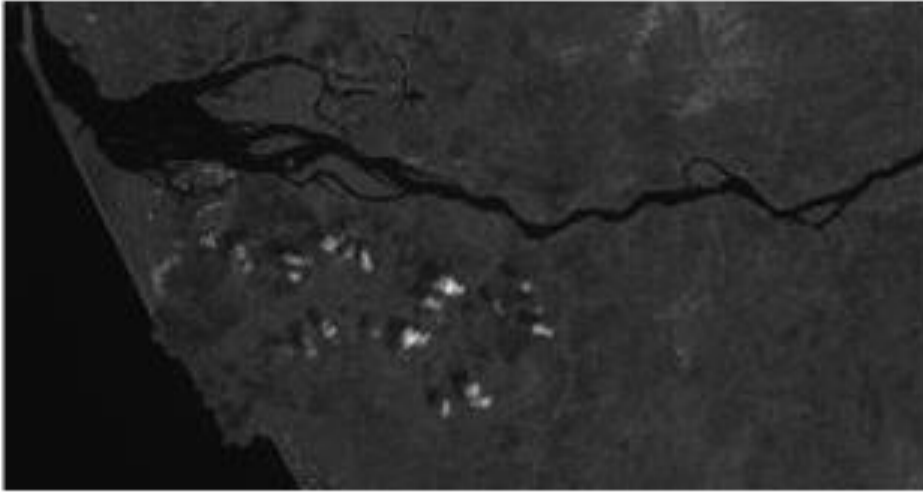
&

$$\text{Sobel gradient} = \sqrt{x^2 + y^2}$$

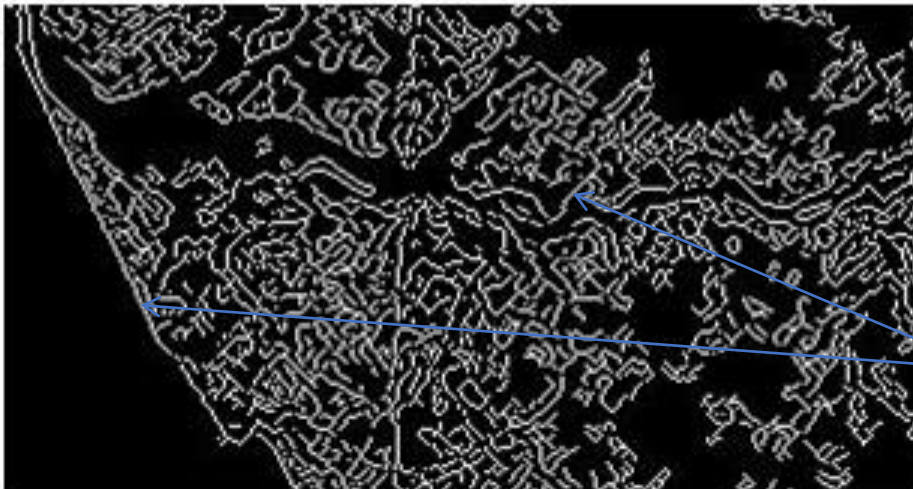
Edge Enhancement

- Edges are generally formed by long linear features
- Example: Ridges, rivers, roads, railways, canals, folds and faults
- Such narrow linear features in the image can be enhanced using appropriate filtering techniques called **Edge Enhancement**

Edge Enhancement ...



IRS LISS III Band 4 (Near Infrared)
data showing a portion of Uttara
Kannada district in Karnataka state,
India



Edges extracted from the image

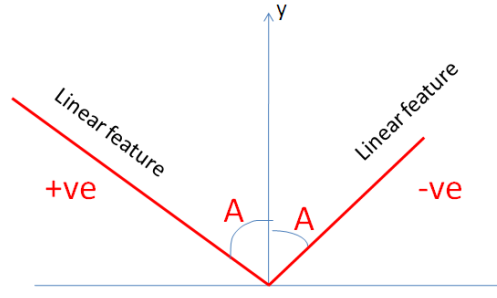
Linear features such as shore line
and the river banks are extracted

Digital Filters for Edge Enhancement

- Digital filters are used to detect the linear features in the image
 - Directional filters : Enhance linear features having specific orientation
 - Non-directional filters : Enhance linear features in almost all directions
 - Example: Prewitt gradient, Sobel and Laplacian filters

Directional Filters

- Enhances features with specific orientation (Eg., 30°NW, 45°NE)
- Orientation (A) is specified with respect to the North
 - +A in the NE direction and -A in the NW direction
- Uses 2 kernels of size 3x3: Left and right kernels



-1	0	1
-1	0	1
-1	0	1

Left Kernel

1	1	1
0	0	0
-1	-1	-1

Right Kernel

Directional Filters ...

- Both left and right kernels are moved over the input image
- Original image pixel values are multiplied by the kernel coefficients and added up for each kernel, separately
- Value added up for the right kernel is multiplied by $\sin(A)$
- Value added up for the left kernel is multiplied by $\cos(A)$
- Resulting two kernel values are summed to obtain the directional filter value

Directional Filters ...

$$\text{Cos } A * \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} + \text{Sin } A * \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

a. Directional Filter

Example Data

50	50	50	50	50	50
50	50	50	50	50	45
50	50	50	50	45	45
50	50	50	45	45	45
50	50	45	45	45	45
50	45	45	45	45	45

$A = -45^\circ$

Sharp gradient indicates linear feature

b. Original Data

0	0	0	0	-7	-14
0	0	0	-7	-14	-14
0	0	-7	-14	-14	-7
0	-7	-14	-14	-7	0
-7	-14	-14	-7	0	0
-14	-14	-7	0	0	0

Absolute value is maximum along the gradient

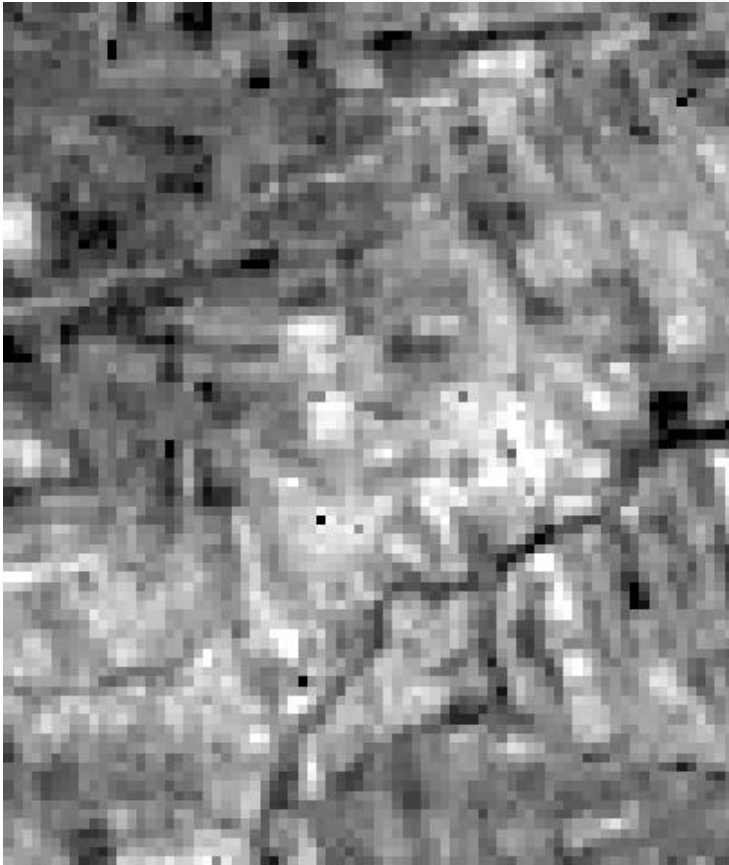
c. Kernel Values

50	50	50	50	43	36
50	50	50	43	36	31
50	50	43	36	31	38
50	43	36	31	38	45
43	36	31	38	45	45
36	31	38	45	45	45

Contrast ratio is enhanced along the edge

d. Filtered Data

Directional Filters ...



Landsat ETM⁺ Band 5 image



Landsat ETM⁺ Band 5 image after
right diagonal edge enhancement

Directional Filters ...



Landsat ETM⁺ Band 5 image after
horizontal edge enhancement



Landsat ETM⁺ Band 5 image after
left diagonal edge enhancement

Non-Directional Filter: Laplacian Filter

- Linear features in an image are identified using the contrast between the pixels
 - Difference in the pixel values ($a_{x+1} - a_x$) = First derivative of the pixel values

						Edge			Noise				Edge			
Pixel (x)		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Value (a)		50	50	40	30	20	10	10	10	60	10	10	10	50	50	60
1st derivative	$a_{x+1} - a_x$		-10	-10	-10	-10	0	0	50	-50	0	0	40	0	10	
2nd derivative	$a_{x-1} + a_{x+1} - 2a_x$		-10	0	0	0	10	0	50	-100	50	0	40	-40	10	

- First derivative delineates broad edges
- Narrow edges are better delineated using the second derivative**
- Noise in the data are also better identified using the second derivative**
- Laplacian filters are based on second derivatives

Laplacian Filter

- A non-directional filter based on the second spatial derivative of the pixel values
- The second order derivative is given as

$$\frac{\partial^2 a}{\partial^2 x^2} = a_{x-1} + a_{x+1} - 2a_x$$

$$\frac{\partial^2 a}{\partial^2 y^2} = a_{y-1} + a_{y+1} - 2a_y$$

$$\nabla^2 a = \frac{\partial^2 a}{\partial^2 x^2} + \frac{\partial^2 a}{\partial^2 y^2} = a_{x-1,y} + a_{x+1,y} + a_{x,y-1} + a_{x,y+1} - 4a_{x,y}$$

	$a_{x,y+1}$	
$a_{x-1,y}$	$a_{x,y}$	$a_{x+1,y}$
	$a_{x,y-1}$	

- This equation is implemented in the Laplace kernel

0	1	0
1	-4	1
0	1	0

	40	40	40	40	35	35	40	40	40	40	40	40	40	45	45	45	
	40	40	40	40	35	35	40	40	40	40	40	40	40	45	45	45	
A	40	40	40	40	35	35	40	40	40	40	40	40	40	45	45	45	E
	40	40	40	40	35	35	40	40	40	40	40	40	40	45	45	45	

Laplacian Filter...

Contrast ratio = $40/35 = 1.14$



Contrast ratio = $45/30 = 1.5$

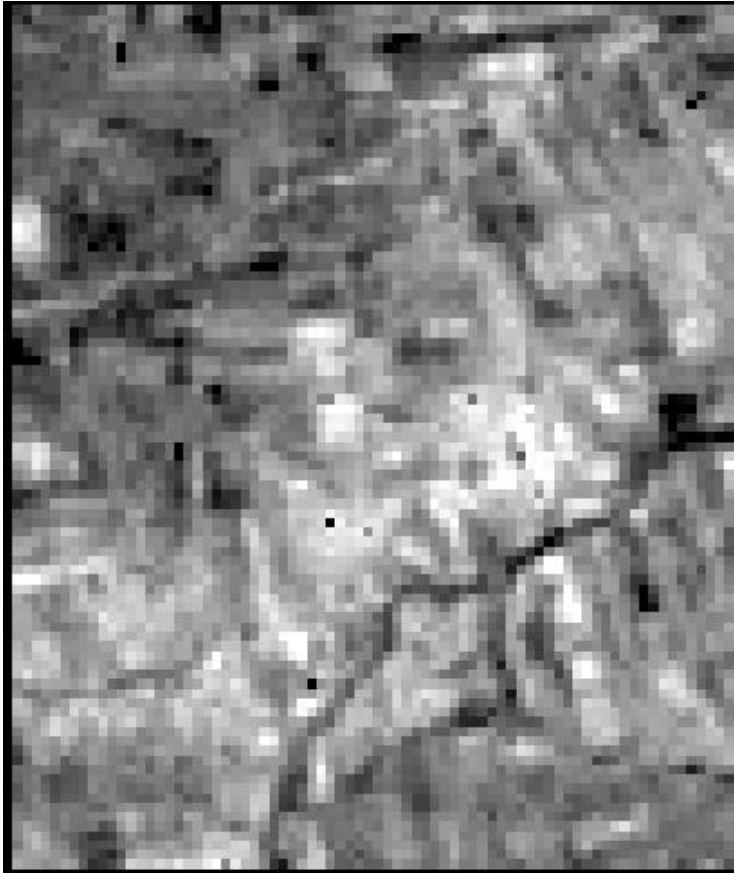
31% increase



Modified from

<http://www.ciesin.org/docs/005-477/005-477.html>

Laplacian Filter...



Landsat ETM+ Band 5 image



Landsat ETM+ Band 5 image after
Laplacian edge enhancement

Laplacian Filter...

- Laplacian filter kernel

0	1	0
1	-4	1
0	1	0

- Having considered the variation in the 4 adjacent directions, the kernel gives isotropic results for rotation in 90° increments
- The 4 diagonals can also be incorporated in the second derivative by adding two more terms for the diagonal corrections.
- Kernel for isotropic results for rotations in 45° increments

1	1	1
1	-8	1
1	1	1

Laplacian Filters...

- Gives isotropic results for rotation in multiples of 45°
- Highly effective in detecting edges irrespective of their orientation
- Mechanics of edge enhancement using Laplacian filter
 - The 3x3 kernel is moved over the input image
 - Pixel values within the kernel mask are multiplied by the kernel coefficients and are added up
 - The resulting value is added to the central pixel to calculate the modified pixel value
 - The procedure is repeated for all the pixels in the input image